
Statistic Preservation Does Not Identify Security

Preservation: Limits of Composed Defenses in Collaborative Learning

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Collaborative learning composes defenses that were never designed together, so
2 upstream processing can destroy what the downstream defense reads. The natural
3 repair is to require the upstream stage to **preserve** it; for the broad class of *pos-*
4 *itive per-client rescaling* compositions, where the invariance question is exactly
5 algebraic, that is not enough. The obstruction is structural: a criterion gated on
6 the outcome it predicts conditions on a descendant of both the mechanism and
7 the defense’s standalone strength, so **no screen of this form identifies the mech-**
8 **anism’s contribution, at any sample size.** Ignoring this does not cost power, it
9 costs the sign: on the identical cell the outcome-gated ladder moves -0.272 where
10 the instrument moves $+0.098$. Preservation is a hierarchy, and we break its last
11 two links by intervening rather than comparing: Oracle intervention pinning every
12 adversary’s coefficient closes the attenuation channel, and disturbing the statistic
13 Krum reads then flips its decision in up to 80% of rounds while the *support* of
14 the admitted adversarial mass never changes in any measured round of any arm,
15 and suppression stays inside our pre-specified ± 0.15 equivalence margin. We
16 prove sufficient conditions for the weighting $\rightarrow$ aggregator class and replicate the
17 negative in a second invariance class and on a second dataset and architecture.
18 The implication for decentralized training is methodological: evaluate composed
19 mechanisms by intervening on adversarial contribution, not by inferring it from
20 downstream statistics.

21 1 Introduction

22 Federated learning [McMahan et al., 2017] is vulnerable to data poisoning [Chen et al., 2017, Bag-
23 dasaryan et al., 2020], and an arms race has produced defenses [Fung et al., 2020, Pillutla et al.,
24 2022, Yin et al., 2018, Blanchard et al., 2017] with accuracy/robustness trade-offs and none univer-
25 sally dominant (Appendix F). *Can we compose them for stronger robustness?* Composition does
26 not even commute: FoolsGold $\rightarrow$ CoordMedian suppresses a pixel backdoor to ASR 0.076 while the
27 reverse order reaches 0.732: a rank aggregator emits one vector rather than per-client updates, so it
28 cannot be an *upstream* stage and the pair collapses to FoolsGold alone, bit-identically. Naive layer-
29 ing fails this way whenever upstream processing destroys what the downstream defense reads, and
30 it bites hardest in decentralized pipelines, foundation models included.

31 **Why the question is sharper without a trusted coordinator.** Such stacks compose independently
32 developed mechanisms among heterogeneous, partially trusted, often pseudonymous participants,

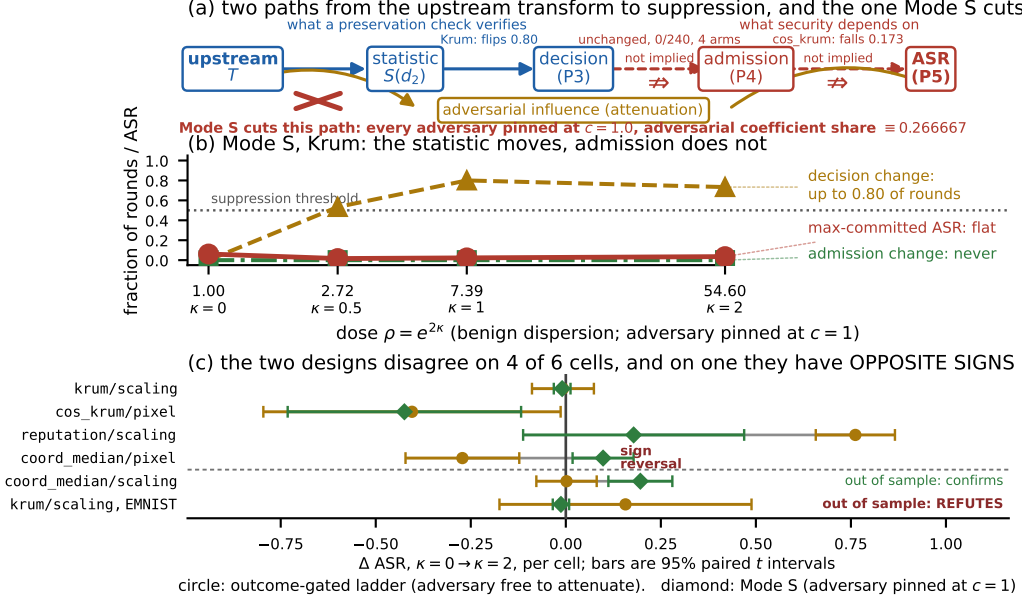


Figure 1: **The paper in one figure: the statistic moves, admission does not, suppression does not follow.** (a) Two paths run from an upstream rescaling T to suppression: the *chain* a preservation criterion reasons along, and a *confounded* path on which T merely attenuates whichever client carries the poison (Lem. 1) and reaches ASR **without passing through d_2 's statistic at all**. **Mode S cuts the second by construction**, pinning every adversary at $c=1$ (the one channel the dose closes, and not the only thing it moves). The admission count pools the four aggregators of §5; Krum admits no adversary at any rung, so its zero is a floor (App. C). (b) Mode S, Krum under model-scaling (§4): as ρ grows to 54.6 the *decision* flips in up to 80% of rounds while the *support* of the admitted adversarial mass does not, and max-committed ASR stays inside the pre-specified ± 0.15 equivalence margin ($|\Delta| = 0.026$). Error bars in this panel are Student- t 95% intervals on the levels ($n=5$). (c) **Six cells, one contrast each, estimated two ways** (circle: outcome-gated ladder; diamond: Mode S, adversary pinned at $c=1$). **The designs disagree on 4 of 6 cells, and on coord_median/pixel they have opposite signs:** $\Delta \text{ASR} = -0.272$ (paired 95% t interval $[-0.422, -0.122]$) against $+0.098$ ($[+0.018, +0.178]$), intervals excluding zero and not overlapping, so **closing the attenuation channel does not shrink the effect, it reverses its sign**. Above the dashed rule are the four cells a pre-registered mechanism was read off (*training data, not evidence for it*); below are its two out-of-sample cells, one of which **refutes it, so we withdraw it**.

so preservation is the obvious discipline to demand of them. Concretely: a contributor clips its own update, a scheduler reweights contributors by reputation, and whoever merges runs a robust aggregator: three parties, three mechanisms, nobody owning the stack. The only check at each seam is that the next stage still reads what it expects, so a preserved statistic is read as evidence the stack is robust. **That inference is what this paper refutes. The identification problem we isolate is defined independently of model scale**, so we study it in a controlled federated setting rather than claiming transfer to foundation-model training; statistic-preservation checks are therefore not security evidence in larger decentralized stacks, and assuming otherwise in a trustless pipeline is worse than uninformative.

Contributions. We propose no new defense, and **our central contribution is a negative identification result**: an outcome-gated comparison of defense pairs cannot determine whether preserving a downstream defense's mechanism is what keeps the composition secure. Everything below is that result's demonstration or its consequence. *Security preservation* names the object throughout: suppression of the committed attack surviving composition, instantiated as ASR, never the statistic or the decision that follows, and we claim nothing about confidentiality, availability, or robustness beyond that suppression; Fig. 1 is the paper in one picture. **The boundary.** Outcome-gated cross-defense screening is structurally unidentified: its eligibility gate is defined on the same out-

come it predicts, so no cross-defense contrast *satisfying this outcome-gated design* identifies the mechanism’s contribution and the admissible designs are within-defense (§3). **The demonstration.** Statistic preservation is accordingly not a valid causal proxy for security preservation, established by intervention rather than correlation: on the identical cell the outcome-gated design estimates -0.272 where the intervention estimates $+0.098$, reversing the direction and not merely the precision (§4). **The consequence.** The screening rule follows from the boundary and is nothing more: evaluation prioritization, measurably broken under heterogeneity (App. J.2). **Also.** The preservation hierarchy (Def. 1); mechanism preservation as invariance under positive per-client rescalings (Prop. 1), with margins on the weighting $\rightarrow$ aggregator class (Thm. 1), supporting machinery, safe at *both* extremes of the weight ratio, so dispersion alone mispredicts; counterexamples across four aggregators of three structural kinds; and the negative replicated in a second invariance class and on a second dataset and architecture, while refuting the cross-arm admission ordering we had predicted. Scope is inventoried once, in Tables 1 and 2: what each claim does *not* license, and the evidence tiers (App. J.1).

2 Mechanism Preservation: Invariance Classes

What does it mean for a composition to preserve a defense? Every per-client defense we consider acts as a positive scalar rescaling $T(u_i) = c_i u_i$, which gives the invariance question a sharp *algebraic* answer (Prop. 1), but writing C2 as an invariance of the statistic S leaves a gap our own experiments exposed. Three different things can be preserved: what the downstream defense *reads* (P1–P2), what it *decides* and what *survives* it (P3–P4), and whether the attack is still *suppressed* (P5). They come apart, so we name the levels and ask of each whether it determines the one below:

Level	What is preserved	Determines the level below?
(P1)	the statistic’s value	yes, by definition
(P2)	the ordering it induces	yes, by definition
(P3)	the decision d_2 makes	no , by counterexample (§4)
(P4)	the adversarial contribution admitted	no , by counterexample (§5)
(P5)	suppression of the committed attack	—

Both breaks are measured, at fixed d_2 and attack: Krum’s decision flips in up to 80% of rounds with the adversarial mass it admits unchanged (§4), and `cos_krum`’s decision and admitted mass are bit-identical while ASR falls 0.173 (§5); Fig. 2 draws the chain and both breaks.

The algebra fixes which level a given composition sits at: cosine similarity is exactly invariant, so C2 holds for every $\cdot \rightarrow$ FoolsGold composition; coordinate ordering survives only under a bounded-ratio condition; consensus distance and residual magnitude are not invariant at all, so $\cdot \rightarrow$ Reputation compositions are C2 failures; norm clipping does not blind FoolsGold, though it does blind Reputation.

Definition 1 (Preservation levels). *For an upstream transform T and a downstream defense d_2 reading statistic S :*

- (P1) **Value invariance:** $S(T(U)) = S(U)$.
- (P2) **Ordering invariance:** $S(T(U)) = \varphi(S(U))$ for some φ preserving the ordering S induces on clients. (this is C2).
- (P3) **Decision invariance:** the client or coordinate d_2 actually selects is unchanged.
- (P4) **Adversarial contribution (admission) invariance:** the adversarial input d_2 admits into its aggregate is unchanged. Admission is the pre-registered term throughout for how much adversarial input survives d_2 : a level, not one measurable quantity, instantiated by whatever each rule exposes: `Krum`’s and `cos_krum`’s selected-client indicator; `coord_median`’s selected-coordinate share, reputation’s adversarial weight share, which is why App. C’s influence column is aggregator-specific.
- (P5) **Attack-suppression preservation:** ASR stays below threshold.

We target (P3) $\nrightarrow$ (P4) in §4 and (P4) $\nrightarrow$ (P5) in §5. The conditions (C0)–(C3) the levels induce (**the rule being $C1 \wedge C2$**) are in App. A and B.

What the theory delivers: supporting machinery, not the claim. For a per-client weighting into a rank-based or geometric aggregator, Thm. 1 gives sufficient conditions for d_2 ’s ordering to

97 survive: it **certifies mechanism preservation, not security** and bounds no ASR. Its one prediction,
 98 admitted adversarial mass $\Lambda_a \leq 12.6\%$ against a realized $\leq 1.8\%$, goes vacuous as the separation
 99 margin approaches 1, which heterogeneity reaches, and does not follow from a (d_1, d_2) definition
 100 pair alone, as a screen would need (App. I).

101 3 The Testability Boundary

102 *Why can we not simply compare many defense pairs?* The natural test keeps pairs whose constituents
 103 already suppress the attack and asks whether the composition preserves it. That is exactly where
 104 identification is lost. **The boundary in words** (Prop. 4, formal in App. J.4): let a screen certify
 105 $(d_1 \rightarrow d_2)$ only if an *eligibility gate* holds: some constituent already suppresses the attack, measured
 106 by the same outcome metric and threshold the screen predicts for the composition, and a mechanism
 107 condition P holds, fixed by the two defenses' definitions alone. Four cells follow, and that design
 108 never fills two of them:

	Mechanism condition	d_2 suppresses a alone	it does not
109	P holds (preserved)	observed: every certified pair	empty by the gate
	$\neg P$ (destroyed)	observed: predicted-FAIL pairs	empty by the gate

110 Reading down a column is the contrast that would identify P 's contribution, and **the only column**
 111 **in which d_2 is not already doing the work is empty**, for every menu a screen of this form could
 112 range over: excluded by design, not undersampled: more data cannot fill a cell the design never
 113 permits. The gate cannot be dropped either, because without it C1 is vacuous. So wherever our
 114 conditions certify a pair, a low composition ASR is attributable to d_2 's standalone strength rather
 115 than to preservation. Equivalently, this is selection on a collider [Elwert and Winship, 2014, Pearl,
 116 2009]: the gate is a descendant of both P and d_2 's standalone effectiveness, so conditioning on it
 117 makes them co-vary. The hazard is standard; the gate reading as hygiene rather than as the step that
 118 removes identification is not, and we made that mistake ourselves (§5).

119 **The admissible design, derived: the practical contribution.** Read forwards the boundary is
 120 *constructive*, each clause forced by a term in Prop. 4: (1) hold d_2 and the attack fixed, so the gate's
 121 input is the same quantity on both sides of every contrast; (2) intervene on the upstream transform
 122 rather than selecting a different d_2 , whose variation is what empties the cell; (3) pin the adversarial
 123 coefficient channel, since an upstream that attenuates the adversary moves the outcome by a route
 124 unrelated to d_2 's statistic, the confound that inverted our own sign; (4) measure every channel:
 125 aggregate displacement, decision, admission, influence. §4 is clause (3) as an instrument; the six-
 126 step form is in App. J.3. A second boundary bounds what such a screen can *find*: **no emergent**
 127 **composition can satisfy an inherited-suppression gate**, our own lowest-ASR composition being
 128 the witness: $FG \rightarrow RFA$ reaches 0.093 where FoolsGold alone reaches 0.732, so the recall bound
 129 binds at $1/5$ (App. J.2).

130 4 Statistic Preservation Does Not Identify Security Preservation

131 *Does disturbing what a defense reads actually cost suppression?* **The claim: it need not, and the**
 132 **design that says otherwise gets the sign backwards.** Our pre-registered pooled hypothesis (H1),
 133 that ASR rises with disturbance of the statistic d_2 reads, is rejected, the effect *negative*: that ladder
 134 moved adversarial influence and the statistic together (§5). We adjudicate C2's implication that
 135 losing (P3) means losing (P5) by oracle intervention on two arms in two invariance classes, not by a
 136 general law.

137 **The instrument, as a counterfactual.** The clean question asks what happens when the down-
 138 stream statistic changes but the adversary cannot become weaker. **Mode S** enforces exactly that: it
 139 pins every adversary's coefficient at $c=1.0$ and spreads only the benign clients over $\rho = e^{2\kappa}$, hold-
 140 ing the adversarial coefficient share constant at 0.266667 to within float32 read-back noise, closing
 141 Figure 1(a)'s attenuation path, so a rise in ASR here cannot be attenuation. Mode S and Mode A (§5)
 142 are causal instruments built from adversary identity: they measure mechanism-level effects rather
 143 than emulate deployable defenses (Table 1, App. J.5).

144 **Two readings of C2, frozen against each other.** *The statistic reading:* what matters is disturbance
 145 of d_2 's statistic, read off its *decisions*. *The admission reading:* what matters is the *adversarial input*
 146 d_2 *admits*. Mode S disturbs Krum's selection heavily while leaving the mass it admits untouched,
 147 so that arm adjudicates (Table 4, the pre-registration's H-statistic and H-admission): as ρ climbs
 148 to 54.6 we find the decision changing in up to 0.80 of rounds while the adversarial mass admitted
 149 changes in none of them, and ASR showing no evidence of the predicted increase, remaining in-
 150 side the pre-specified equivalence margin ($p_{\uparrow}=0.70$, $|\Delta|=0.026 < 0.15$). **The statistic reading is**
 151 **refuted:** statistic and decision disturbance need not change suppression once the adversarial coeffi-
 152 cient share is held fixed. *On this arm the zero is a floor:* Krum admits no adversary at any rung, so
 153 the dissociation is read across all four aggregators, each under its own committed attack:

Aggregator	Kind	Δ dec.	Δ adm.	$\Delta\Lambda_a$	Δ ASR
Krum	selector	0.733	0.000	0.000	-0.026
Reputation	weighted averager	1.000	0.000	0.020	+0.178
Cosine-Krum	cosine-invariant selector	0.000	0.000	0.000	-0.425
Coord. median	coordinate order statistic	0.482	0.000	0.033	+0.098

155 Δ adm. is the *support* of the admitted adversarial mass: it never moves in any of the 60 measured
 156 rounds of any arm, including the 48 per arm where reputation and coord_median carry adversarial
 157 mass at baseline. $\Delta\Lambda_a$ is the separate question of how much weight the adversary's direction carries.
 158 cos_krum's -0.425 is the full-ladder endpoint, below our accuracy floor; the witness we rest on is
 159 its first, gate-passing rung (0.173, §5); App. C adds EMNIST-byclass and the score-only control.

160 **What the instrument does not hold fixed, and the pre-registered control that closes it.** Mode S
 161 pins the *adversarial* coefficients and nothing else: the dose also moves Krum's pairwise distances,
 162 *which* benign client it selects, and the **magnitude and geometry of the update actually aggre-**
 163 **gated.** That third channel is measured: App. C's Δ agg. column, the aggregate the defense *emits*,
 164 reads **0.892** here (App. D). **Score-only Krum** closes it: Krum scores the transformed stack and so
 165 makes the same selection, but the *untransformed* selected update is aggregated: the decision channel
 166 preserved, the magnitude and geometry change removed. Rules were frozen at 35788d9 before any
 167 score-only ASR existed, $\Delta > +0.15$ meaning the published result was partly a magnitude artifact.
 168 We find $\Delta = -0.023$ ($0.062 \rightarrow 0.039$) against -0.026 uncontrolled on the identical cell, at the
 169 same Δ dec. of 0.733, the refuting threshold 0.212 nowhere approached and every rung clearing the
 170 accuracy floor, so the central claim is instrumented against two independent confounds, attenuation
 171 and magnitude (App. E).

172 **A pre-registered replication that refutes the positive half, and the sign reversal.**
 173 coord_median under the pixel backdoor is *rank-discordant*: of the interpretable arms it has the
 174 smallest decision change (0.482) and the largest admission-level change ($\Delta\Lambda_a=0.033$), so the two
 175 readings predict opposite rank positions for its rise, and thresholds were frozen against the two pub-
 176 lished rises (confirm if $\Delta > +0.178$, refute if $\Delta < +0.150$). We find $\Delta = +0.098$, which **refutes**
 177 **the admission ordering:** the arm with the largest admission-level change rose *less* than one with a
 178 smaller change. The decision reading does not predict it either ($R^2=0.239$ pooled). What replicates
 179 is the negative on a coordinate-wise order statistic: a decision change of 0.482 costs no suppression
 180 by the frozen margin, at accuracy 0.698. Identity ASR here is 0.443, so equivalence is to the *identity*
 181 *run*; the rise is positive in 5/5 seeds (post-hoc): small and consistent, not zero.

182 **The confounded design gets the causal sign wrong.** On this identical cell (matched displace-
 183 ment, decision change and clean accuracy, Table 3), closing one channel reverses the sign:

Design	The adversarial coefficient	Δ ASR
Outcome-gated ladder (§5)	free to attenuate	-0.272 ($p_{\downarrow}=0.0015$)
Mode S intervention (§4)	pinned at $c=1$	+0.098

185 **The confounded design does not lose power, it gets the sign wrong.** Closing the attenuation chan-
 186 nel inverts the effect rather than sharpening it: a criterion calibrated against the confounded contrast
 187 would have been calibrated against the confound, so §3's boundary is not a technicality about power.
 188 **Downstream-statistic preservation is not, in general, sufficient to establish attack-suppression**
 189 **preservation** (Prop. 2): no predicate of the statistic alone determines the admitted adversarial mass,
 190 and none of that mass alone determines suppression: the title claim, by counterexample *within this*
 191 *class* (Table 1).

5 What the Boundary Licenses, and What We Found

Does the pattern hold beyond the adjudicating arm, and what does a screen then buy? We validate on CIFAR-10 with a 4-layer CNN, $N=10$ clients, $f=0.2$ adversarial, two committed attacks (model-scaling, pixel backdoor), at $n=5$ seeds per cell ($n=8$ for `cos_krum`, $n=3$ on EMNIST-byclass); ASR means *max-committed ASR*; the rest of the protocol is fixed in App. X. The core evidence is a seed-matched intervention on one cell, not a mean comparison across cells: what §3 excludes is a cell, not a sample size.

The confounded ladder that motivated the instrument, and the sign it got wrong. Our earlier suites tried to validate C2 by *choosing a different downstream defense*, the move Prop. 4 rules out. Varying only the *dose* at fixed d_2 and attack ($\kappa=0$ bit-identical to d_2 standalone) refutes that suite’s pooled prediction that disturbance predicts each arm’s rise: $\rho_s = +0.351$, one-sided $p=0.091$ against a frozen $p < 0.05$, and in three of four arms ASR *falls* as the dose rises (80 runs, one arm confirming decisively; Table 5). **The fault is ours and is now measured:** fixing $\text{mean}(c)=1$ leaves relative weights free, so a large spread mostly *dilutes* whichever client carries the poison: attenuation, not statistic disturbance (App. M, R).

What the instrument does to that ladder, across four aggregators and a second dataset. Most of the one confirmation was attenuation: with the adversary’s share pinned, reputation’s rise at $\rho=54.6$ shrinks from $+0.762$ to $+0.178$, and what remains tracks a small *admission* change, not the large decision change (App. J.5). The class-(a) control refutes us the other way: `cos_krum`’s selection is bit-identical at every rung yet ASR is 0.584 at the identity (it never suppresses) and *falls* 0.173 at the first accuracy-gate-passing rung (one-sided $p=0.044$, 5/8 seeds); the larger fall across the full ladder sits below our own accuracy floor, so **the gate-passing rung is the witness we rest on.** **Mode A** (payload share at fixed dispersion) separates Thm. 1’s two regimes: confirmed for `coord_median`, refuted for reputation. A *pre-registered* replication on EMNIST-byclass/simple_cnn moves the decision on 0.643–0.714 of rounds with the adversary never admitted, suppression inside the pre-specified ± 0.15 equivalence margin: $\Delta = -0.017$ beside CIFAR-10’s -0.026 : **replicated** on a second dataset and architecture, never “generalizes” (App. C, K).

And what a screen then buys. Only the 13/20 prospective sub-tier risks falsification (Table 7), and what limits it is a property of our menu rather than of the screen: FLTrust suppresses both attacks alone (0.048), so its pairs are out of scope by C0 or near-identities. Two pre-registered attempts to validate the *screen* were defeated by that same confound, which makes them evidence for Prop. 4. **Scale:** the pre-registered PASS predictions hold at $N=100/\text{ResNet18}$ but far from converged accuracy, so they bound nothing about converged training. Under *one* adaptive stress test $\text{FG} \rightarrow \text{RFA}$ rises $3.3\times$: **inherited robustness is not adaptive robustness** (App. K, O, S).

6 Conclusion

The finding. Downstream-statistic preservation is not sufficient to identify preserved attack suppression. An intervention changes a defense’s statistic and decision substantially without changing the adversarial mass it admits or the suppression that follows; conversely a bit-identical statistic and decision coexist with a change in suppression, each direction with a measured witness.

Why. An outcome-gated cross-defense comparison conditions on the same outcome it seeks to explain, so it cannot separate mechanism preservation from standalone effectiveness: a structural identification boundary (Prop. 4) no amount of additional data within that design removes. Within-defense intervention fixes d_2 and the attack instead: on the identical cell **the confounded design does not lose power, it gets the sign wrong.**

The implication. Evaluate composed defenses by measuring the adversarial contribution that survives the downstream mechanism, not by inferring security from the statistic it consumes. That object is (P4), which Mode S reads only by knowing adversary identity, so we identify the channel rather than screen for it (App. J.3). The open question: **can (P4) be estimated without knowing who is malicious?**

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Table 1 sits ahead of the appendices because it governs all of them: it is the one place each scope condition is stated, and no later section restates them at each use.

Table 1: **Scope of claims.** The one place each scope condition is stated. **Claims do not transfer upward:** the formal guarantee does not certify the general framing, and the empirical figures certify neither level above them.

kind	content	status, and what it does <i>not</i> license
General	An evaluation gated on the outcome it predicts cannot identify the mechanism’s contribution; admissible designs are within-defense (Prop. 4, Prop. 5).	Proved, and not specific to FL, to backdoors, or to our conditions. Licenses no claim that a composition <i>is</i> secure, nor that mechanism preservation implies suppression.
Formal	Mechanism preservation as <i>invariance</i> under positive per-client rescalings $T(u_i) = c_i u_i$, $c_i > 0$, into specific rank and geometric aggregators (Thm. 1, Lem. 1, Cor. 1).	Proved on that class only: a <i>laboratory</i> for composition, not a model of FL defenses in general. A d_1 that alters participation, carries cross-round state, mixes clients, or aggregates first has no such T and is covered by nothing here (C0, applied retrospectively to arms frozen before it). Bounds no ASR , and Regime A’s condition is not operationally checkable (S is not a stable per-attack constant).
Empirical	§4’s interventions and §5’s 42-pair menu: CIFAR-10, CifarCNN/ResNet18, $N \in \{10, 100\}$, $f \in [0.05, 0.6]$, Dirichlet $\alpha=0.5$, star topology , three attack families; one arm on EMNIST-byclass/simple.cnn.	Measured at $n=5$ seeds per cell ($n=8$ for <code>cos_krum</code> , $n=3$ on EMNIST-byclass), fixed before the runs, so intervals are wide and no null is read as flatness without a pre-registered equivalence margin. Dataset transfer is tested on the flagship arm alone: “replicated on a second dataset and architecture,” never “generalizes”. “Refuted” always means a pre-registered <i>directional</i> hypothesis was rejected against a rule frozen for it, never that a mathematical claim was disproved.
Both instruments are oracles	Mode S and Mode A build c_i from adversary identity, as does Δ_c .	They identify a causal channel and are not screens ; computing (P4) without adversary identity is our open problem, and admission is <i>a direction to measure</i> , not a validated predictor. C1 and all built on it assume a committed adversary: what is bounded is <i>inherited</i> robustness under a fixed attack, never adaptive robustness .

A The composition model and the invariance classification

The two witnesses under Definition 1, and which rung each rests on. (P1) $\Rightarrow$ (P2) $\Rightarrow$ (P3) hold by definition: logical implications, not empirical claims; their content is that they stop at (P3), and §2’s table states both breaks below it. *Holding* decision and admission invariance does not deliver suppression: `cos_krum` falls 0.173 by the first rung alone through the magnitude channel, at $\rho=2.72$, where clean accuracy moves only $0.599 \rightarrow 0.567$ and our accuracy gate passes (Appendix N). Across the whole ladder it falls 0.425, but that endpoint sits below our accuracy floor, so **the gate-passing rung is the witness we rest on** (§5). *Losing* decision invariance does not cost suppression: Krum’s selection flips in up to 80% of rounds under an intervention pinning the adversarial mass, and its ASR stays inside the pre-specified equivalence margin ($|\Delta|=0.026 < 0.15$) (§4). So a criterion stated at (P1)–(P2) constrains security in *neither* direction, **(P4), where the security content sits, is not sufficient for (P5) either, and the hierarchy answers “which level does a security claim have to be stated at?”, not “which level predicts ASR?”**

Definition 2 (Defense composition). *Given defenses d_1 and d_2 , composition ($d_1 \rightarrow d_2$) operates as: d_1 produces per-client transformed updates $U' = \{T_{d_1}(u_i)\}_{i=1}^K$; d_2 aggregates U' using its standard algorithm. For FoolsGold $\rightarrow$ CoordMedian, FG outputs $\{w_i u_i\}$ and CoordMedian takes the unweighted coordinate-wise median of the K transformed values, not a “weighted median”. Clients with $w_i = 0$ are **not removed** but reach d_2 as zero vectors, so at least one point of U' is always the origin, which is what Lemma 1 is about.*

Each defense suppresses malicious updates through a *discriminative property* (a structural feature separating adversarial from benign contributions) and the property is defense-class-specific: *similarity spread* (FoolsGold), *coordinate ordering* (TrimmedMean, CoordMedian), *residual magnitude* (RFA), *consensus distance* (Reputation, in ℓ_2 from the coordinate-wise median) and *norm magnitude* (NormClip). Mechanism preservation is then governed by whether d_1 ’s transform leaves d_2 ’s property intact, and because every per-client defense we consider acts as a positive scalar rescaling, that question has a sharp answer:

Proposition 1 (Invariance classes under positive per-client rescaling). *Let $T(u_i) = c_i u_i$ with $c_i > 0$, strictly: the form taken by NormClip ($c_i = \min(1, \tau/\|u_i\|)$), Reputation ($c_i \propto e^{-d_i/s}$) and RFA-reweighting ($c_i \propto 1/\max(\|u_i - \mu'\|, \varepsilon)$), each positive by construction. FoolsGold is deliberately*

323 *not in that list: its shift-and-clamp gives the least-trusted client weight exactly 0 in every round (2 of*
 324 *5 clients here), so FoolsGold-as- d_1 falls outside the hypothesis and is governed by Lemma 1 rather*
 325 *than by (a)–(c). Then:*

- 326 (a) **Cosine similarity is exactly invariant:** $\cos(c_i u_i, c_j u_j) = \cos(u_i, u_j)$. Hence FoolsGold’s
 327 weight vector is unchanged, $w(T(U)) = w(U)$, and **C2 holds for every $\cdot \rightarrow$ FoolsGold**
 328 **composition.**
- 329 (b) **Coordinate ordering is not invariant** in general; it survives under the bounded-ratio con-
 330 dition of Theorem 1(1).
- 331 (c) **Consensus distance and residual magnitude are not invariant:** heterogeneous c_i can in-
 332 vert their ordering, so $\cdot \rightarrow$ Reputation compositions are C2 failures and $\cdot \rightarrow$ RFA requires
 333 the separation condition of Theorem 1(2).

334 So mechanism preservation is a question about the *invariance class* of the downstream signal, and
 335 the classification corrects a natural but wrong intuition: norm clipping does not blind FoolsGold,
 336 though it does blind Reputation, whose signal is consensus distance rather than similarity.

337 **The framework in general form.** The structure does not depend on rescaling. Write d_2 ’s dis-
 338 criminative property as a statistic $S(U)$ and let T be d_1 ’s per-client transform; then $(d_1 \rightarrow d_2)$ is
 339 **mechanism-preserving** when $S(T(U)) = \varphi(S(U))$ for some φ preserving the ordering d_2 acts on.
 340 C2 is exactly that requirement, C3 asks in addition that φ not compress the gap below d_2 ’s operating
 341 threshold, and Proposition 1 is the case $T(u_i) = c_i u_i$ (Appendix I), so the conditions are alge-
 342 braic rather than experimental. It also states the real limit: **a d_1 whose transform is not per-client**
 343 **(mixing updates, or carrying cross-round state) has no T of this form, and nothing here applies.**

344 **The full causal chain.** Definition 1 is a prefix of a longer one: statistic $\rightarrow$ decision $\rightarrow$ admission $\rightarrow$
 345 aggregate adversarial influence $\rightarrow$ per-round update $\rightarrow$ persistence $\rightarrow$ ASR. **We measure the four**
 346 **emphasized links, not update geometry or persistence dynamics:** an acknowledged gap, not an
 347 assumed-benign one. A selection flipping from one benign client to another disturbs the statistic
 348 maximally and the mechanism not at all, which is why (P3) and (P4) come apart. **“Mechanism”**
 349 **always means the downstream defense’s discriminative operation, never its security behaviour,**
 350 that implication being what this paper refutes.

351 B The signal-preservation conditions in full

352 **Definition 3** (Signal-Preservation Conditions). *Composition $(d_1 \rightarrow d_2)$ is predicted to suppress*
 353 *attack a when: (C0) d_2 ’s mechanism is load-bearing, $\text{ASR}(d_1, a) \geq 0.5$ for some committed a : a*
 354 *condition on the criterion’s domain that **never certifies a pair**, and one we formulated only after*
 355 *§5 made it bind, so it is applied **retrospectively** and is **not part of what the 42 pairs validate**; (C1)*
 356 *some constituent suppresses every committed attack alone, $\min_{d \in \{d_1, d_2\}} \text{ASR}(d, a) < 0.5$ for all*
 357 *a , defined through the same threshold that constitutes the outcome, hence an **empirical prerequisite***
 358 ***and not a mechanistic property**; (C2) d_1 ’s transform does not destroy d_2 ’s discriminative property,*
 359 *decided from definitions alone by Proposition 1; (C3) it does not compress the quantitative margin*
 360 *below d_2 ’s operating threshold. **The prioritization rule is $C1 \wedge C2$, C3 demoted to a margin diag-***
 361 *nostic (Appendix U); the pre-registered labels of §5 were frozen with C3 in the gate and are not*
 362 *relabelled. **Theorem 1 bounds no ASR: that mechanism preservation plus C1 predicts low ASR is a***
 363 *separate empirical hypothesis, tested in §5, and **neither stage alone establishes low ASR, nor does***
 364 ***their conjunction** (stage structure, Appendix B).*

365 Definition 3 in the body states the four conditions and every disclosure attached to them; what
 366 follows is the same criterion with its *stage structure*, which is worth separating because conflating
 367 the stages is the main source of confusion about what the framework claims.

368 Composition $(d_1 \rightarrow d_2)$ suppresses attack a when the conditions below hold. The criterion is a **two-**
 369 **stage screen**, and we label the stages because conflating them is the main source of confusion about
 370 what the framework claims: stage 1 is empirical screening of the *constituents*, stage 2 is mechanistic
 371 screening of the *composition*.

372 **Stage 0: applicability (C0).** Whether the framework has anything to say about this pair at all. This
 373 is a condition on the criterion’s *domain*, and Lemma 2 is stated over it together with C1.

374 **(C0) The downstream mechanism is load-bearing.** $\text{ASR}(d_1, a) \geq 0.5$ for at least one committed
 375 attack a . C2 and C3 concern preservation of d_2 's mechanism, so they carry information only when
 376 that mechanism does work the composition needs: if d_1 alone suppresses everything, the composi-
 377 tion inherits that suppression and a correct prediction is correct for a reason the framework did not
 378 supply. The asymmetry is deliberate: a strong d_2 is the intended case. **Failing C0 removes a pair**
 379 **from the criterion's domain; it never certifies one.** *Provenance:* C0 was vacuous on the menu
 380 the framework was developed on (strongest single defense 0.519) and was formulated only after the
 381 prospective suite of §5 made it bind, so it is applied **retrospectively** and is **not part of what the 42**
 382 **pairs validate** (Appendix Q).

383 **Stage 1: empirical constituent screening** (C1). Read off standalone evaluations. This screens; it
 384 does not explain, and it is not a mechanistic claim.

385 **(C1) Individual effectiveness.** For *every* committed attack a in the evaluated set,
 386 $\min_{d \in \{d_1, d_2\}} \text{ASR}(d, a) < 0.5$. The quantifier matters: our headline metric is the *maximum*
 387 over attacks, so requiring a suppressing constituent for only one attack would license compositions
 388 that fail on the other. C1 is defined through the same ASR threshold that constitutes the outcome,
 389 so it is an **empirical prerequisite, not a mechanistic property**: the one condition that cannot be
 390 checked without running something.

391 **Stage 2: mechanistic composition screening** (C2, with C3 as a diagnostic). Determined by d_1 's
 392 and d_2 's definitions alone, no experiment required, and what Theorem 1 proves. **The evaluation-**
 393 **prioritization criterion is $C1 \wedge C2$,** with C3 retained as a **margin diagnostic** rather than a gate for
 394 the reasons scored in Appendix U. *The pre-registered labels of §5 were frozen with C3 in the gate*
 395 *and are not relabelled,* and the demotion changes no reported number.

396 **(C2) Discriminative-property preservation.** d_1 's transform does not destroy d_2 's discriminative
 397 property, as determined by Proposition 1: automatic for cosine-similarity downstream defenses, con-
 398 ditional for coordinate ordering, and violated for consensus distance under heterogeneous rescaling.

399 **(C3) Margin preservation.** d_1 does not compress the quantitative margin below d_2 's operating
 400 threshold. For rank aggregators: extremeness gap survives reweighting. For geometric aggrega-
 401 tors: separation condition holds on transformed points.

402 **What is proved and what is hypothesised.** Stage 1 is empirical and outcome-defined, stage 2
 403 mechanistic, and **neither alone establishes low ASR, nor does their conjunction**: Theorem 1 es-
 404 tablishes that $C2 \wedge C3$ preserve d_2 's discriminative mechanism on the reweighted inputs and **bounds**
 405 **no ASR**. That mechanism preservation plus C1 predicts *low ASR* is a separate empirical hypothesis,
 406 tested in §5.

407 C The four channels, all four aggregators

408 Endpoint change ($\kappa=0$ against $\kappa=2$) in each measured channel under Mode S, for every
 409 downstream defense in the paper. **The first column, Δ agg., is the relative displace-**
 410 **ment of the aggregate the defense emits,** $\|\text{agg}(T(U)) - \text{agg}(U)\|/\|\text{agg}(U)\|$: what enters
 411 training, and **not** the value of a scoring statistic. The measured channels are therefore ag-
 412 gregate displacement, decision, admission and influence. **Recomputed, not transcribed:**
 413 every cell is emitted by `experiments/build_channel_table.py` from `admission_`
 414 `measurement.json`, `femnist_admission.json`, `targeted_dose/summary.json`, `dose_`
 415 `replication/summary.json` and `dose_femnist/summary.json` under `results/`, which
 416 the script only reads; it also asserts that the identity rung displaces nothing (0.00e+00 for all
 417 six rows). **The sixth row is the score-only control** (Appendix E): the first row's arm with the
 418 magnitude channel closed, so its aggregate column is the re-selection component alone (from
 419 `displacement_decomposition.json`) and its decision, admission and influence columns are
 420 the first row's own numbers, score-only leaving the selection untouched; its ASR comes from
 421 `score_only/summary.json`, which the emitter refuses to merge into the ASR sources it shares
 422 cell keys with, so it cannot shadow the first row's published number. The influence column is
 423 **aggregator-specific by necessity** (a selector has no weights) and its instantiation is: adversarial
 424 share of the Weiszfeld/reputation weight mass for the averager, adversarial share of selected
 425 coordinates for the order statistic, and the indicator that an adversarial update is selected at all for

the two selectors. Earlier drafts labelled this arm “FEMNIST” in the tables while the prose said EMNIST-byclass: the loader reads torchvision EMNIST(split="byclass") under a feminist key and partitions it by the label-Dirichlet split used throughout, *not* LEAF’s by-writer split, so the tables now read EMNIST-byclass while the frozen artifact and pre-registration filenames keep the feminist key. The EMNIST-byclass row is the second-dataset replication of §5; its channels were measured prospectively, before its ASR existed.

Aggregator	Kind	Δ agg.	Δ dec.	Δ adm.	$\Delta\Lambda_a$	Δ ASR
Krum	selector (pairwise distance)	0.892	0.733	0.000	0.000	-0.026
Reputation	weighted averager (cross-round state)	0.659	1.000	0.000	0.020	+0.178
Cosine-Krum	selector (cosine distance)	1.297	0.000	0.000	0.000	-0.425
Coord. median	coordinate-wise order statistic	0.670	0.482	0.000	0.033	+0.098
Krum (EMNIST-byclass)	selector (pairwise distance)	0.842	0.643	0.000	0.000	-0.017
Krum, score-only	selector, magnitude channel closed	0.986	0.733	0.000	0.000	-0.023

What this table does and does not show. It shows that the dissociation between the decision channel and the admission channel is not a property of one aggregator: it holds for a selector, for a cosine-invariant selector, for a coordinate-wise order statistic and for a weighted averager, on two datasets. It does **not** order the arms ($\Delta\Lambda_a$ and Δ ASR disagree in rank, which is the refutation of §4 stated in one row-pair) and it is *not* an out-of-sample test: every row is a Mode-S ladder reported elsewhere in this paper.

D Decomposing the emitted displacement: re-selection against rescaling

The Δ agg. column above mixes two things the causal claim needs apart. For a selector, $\text{agg}(U) = u_{s_0}$ and $\text{agg}(T(U)) = c_{s_1} u_{s_1}$, with s_0 the client selected on the raw stack and s_1 the client selected on the transformed one, so exactly

$$c_{s_1} u_{s_1} - u_{s_0} = \underbrace{(u_{s_1} - u_{s_0})}_{\text{re-selection}} + \underbrace{(c_{s_1} - 1) u_{s_1}}_{\text{rescaling}},$$

both components reported relative to $\|u_{s_0}\|$. They are **not orthogonal** (the measured angle runs from $\cos = -0.99$ to -0.43 , mean ≈ -0.65), so their norms need not sum to the total, and the table prints the angle rather than leaving that discrepancy unexplained. What is asserted instead is the exact vector identity: worst residual $2.66\text{e-}07$ over all 240 (arm, family, rung, seed, round) cells, float32 noise. **This decomposes the published number, not a second similar one:** every recomputed total agrees with the corresponding Δ agg. value in `admission_measurement.json` to $0.00\text{e}+00$, on the same seeds, rounds and configuration. Selectors only: a weighted averager and a coordinate-wise order statistic emit no single selected client, so the split is undefined for them. `experiments/measure_displacement_decomposition.py` trains no ASR, writes a new file, and leaves every frozen artifact byte-identical.

Design	Selector	κ	total	re-selection	rescaling	cos	sel. chg.
Confounded ladder	krum	0	0.000	0.000	0.000	—	0.000
		0.5	0.637	0.685	0.333	-0.65	0.600
		1	0.731	0.851	0.685	-0.66	0.733
		2	0.880	1.675	1.674	-0.72	0.867
Instrument (Mode S)	krum	0	0.000	0.000	0.000	—	0.000
		0.5	0.663	0.646	0.308	-0.64	0.533
		1	0.847	1.075	0.666	-0.69	0.800
		2	0.892	0.986	0.973	-0.66	0.733
Confounded ladder	cos_krum	0	0.000	0.000	0.000	—	0.000
		0.5	0.270	0.000	0.270	—	0.000
		1	0.515	0.000	0.515	—	0.000
		2	0.834	0.000	0.834	—	0.000
Instrument (Mode S)	cos_krum	0	0.000	0.000	0.000	—	0.000
		0.5	0.395	0.000	0.395	—	0.000
		1	0.754	0.000	0.754	—	0.000
		2	1.297	0.000	1.297	—	0.000

454 **What it says, and what it does not.** Mode S into Krum at $\kappa=2$: the published 0.892 is 0.986
455 of re-selection and 0.973 of rescaling, so the dose moves *which* benign client is aggregated and
456 *how large* it arrives, in comparable measure. Of the 11/15 rounds whose selection changes there,
457 **none** lands on an adversary: the quantitative form of the claim that Krum’s decision changes are
458 benign-to-benign, and the reason the support of the admitted adversarial mass changes in **none of**
459 **the 60 measured rounds** while the decision channel moves by 0.733, on this arm no adversary is
460 admitted at *any* rung, so that count is a floor, and the arms carrying adversarial mass at baseline are
461 reputation and `coord_median` (48 of 60 rounds each, App. C). `cos_krum` is the informative contrast:
462 its statistic is exactly invariant to positive rescaling, so 0/15 selections change at every rung of *both*
463 designs and its entire displacement (1.297 under the instrument, 0.834 under the ladder) is rescaling.
464 Its fall therefore cannot travel through the decision channel at all, which corroborates §5’s attribution
465 of that fall to the magnitude channel from a quantity measured independently of any ASR. **This is**
466 **a decomposition of a displacement, not of an ASR effect:** it says what the dose moves in the
467 update entering training and makes no claim about suppression, and it is an analysis of runs already
468 reported rather than a new experiment.

469 E The score-only control in full

470 §4 reports the primary outcome: $\Delta = -0.023$ (mean ASR $0.062 \rightarrow 0.039$, seeds 42–46) against
471 -0.026 for the uncontrolled arm on the identical cell, inside the frozen ± 0.15 equivalence margin,
472 with the accuracy floor cleared at every rung (0.565/0.629/0.624/0.630). The verdict wording is
473 the pre-registration’s own for that branch: *the negative survives with the magnitude channel closed*.

474 **What the arm is.** Krum scores on the transformed stack (statistic and decision disturbed ex-
475 actly as in the uncontrolled arm, since the scoring input is the same) and aggregates the *untrans-*
476 *formed* selected update, so what enters training is bit-for-bit what a client produced. It is the
477 reviewer-suggested control in its own terms: the selected update’s magnitude and direction are
478 held fixed while the selection statistic is perturbed independently. Implementation is a defaulted
479 `score_only=False` kwarg, so no existing call site changes and every frozen computation stays
480 bit-identical; the suite’s own harness check verifies that on the published arms. The $\kappa=0$ rung is
481 *imported* rather than re-run, because at the identity the transform returns the stack unwrapped and
482 score-only Krum *is* Krum alone: an exact claim, verified bit-identically at one seed before the arm
483 started.

484 **Three disclosures, from the freeze rather than from hindsight.** (i) The secondary Jonckheere–
485 Terpstra test was specified to detect a monotone *rise* (the outcome that would have refuted flatness
486 with attenuation already closed by construction) and the observed trend is a *fall* ($J=58.5$, $z=-$
487 1.110 , $p_{\uparrow}=0.867$, permutation $p_{\uparrow}=0.874$). We report it and do **not** score it as support for flatness; the
488 pre-registration says in advance that at $n=5$ a null here is not evidence of flatness. (ii) The branches
489 had unequal headroom. With the identity rung at 0.062, refuting us required mean $\text{ASR}(\kappa=2) >$
490 0.212 against a ceiling of 1 and was fully reachable, whereas the indeterminate branch ($\Delta < -0.15$)
491 was *arithmetically unreachable*: ASR cannot fall below 0, so the largest possible fall on this cell is
492 0.062. (iii) **Selectors only.** A weighted averager and a coordinate-wise order statistic emit no single
493 selected client, so “the selected update” is undefined for them and the control does not cover them:
494 the `coord_median` arm carrying the $+0.098$ rise is therefore **not** magnitude-controlled, and neither
495 is reputation’s $+0.178$. Score-only Krum is also an instrument and not a defense: it aggregates an
496 update its own scoring stage never saw. And closing the magnitude channel still does not hold the
497 model trajectory fixed, because which client is selected continues to change, by design, that being
498 the channel left open.

499 **The `cos_krum` corollary, and exactly what it settles.** Section 5 of the same pre-registration as-
500 serted, before the run, that score-only `cos_krum` must be *bit-identical* across rungs: its statistic is
501 exactly invariant under positive rescaling, so the selection cannot move, and with the emitted update
502 untransformed the whole trajectory must repeat. It does (accuracy 0.528400 and ASR 0.003111 at
503 all four rungs, $|\Delta| = 0.00\text{e}+00$ throughout) and the $\kappa=0$ value reproduces the published cell at that
504 seed exactly, so the two pipelines agree at 0.000000 rather than approximately. **What the check**
505 **does not do is confirm the arm’s published fall.** It runs one seed, as frozen, and that seed is un-
506 representative of the arm: under the uncontrolled dose its own ladder *rises* $+0.224$ where the eight-
507 seed mean *falls* -0.425 . Closing the magnitude channel removes all 0.224 of that seed’s movement,
508 which corroborates the *channel*, for `cos_krum` the emitted displacement is entirely rescaling (Ap-

pendix D), so no other route is open, but the attribution of the arm’s fall rests on that decomposition rather than on this check, and settling it empirically would need the arm’s other seven seeds. Expanding a pre-registered seed count after seeing the data is what pre-registration exists to prevent, so we report the invariance as confirmed and the arm-level attribution as **corroborated mechanically**, not established at the arm mean.

F Related work

The contrast in one sentence. Prior work asks whether a defense withstands an attack. We ask whether an *evaluation criterion* can causally attribute the security of a composition to preservation of the downstream mechanism.

Byzantine-robust aggregation, studied in isolation. Krum [Blanchard et al., 2017], Trimmed Mean [Yin et al., 2018], RFA [Pillutla et al., 2022], FoolsGold [Fung et al., 2020], FLTrust [Cao et al., 2021] and FLAME [Nguyen et al., 2022] are each evaluated as a single stage; we propose no new defense and reason about *ordered pairs* of existing ones. Ensemble adversarial training [Tramèr et al., 2018] *assumes* upstream processing preserves downstream signals; we make that assumption explicit, checkable, and (as it turns out) **the wrong abstraction for security**.

Adaptive attacks are why an inherited-robustness guarantee is not a security guarantee. Optimized model poisoning defeats Krum, trimmed mean and FoolsGold once the attack is chosen *against* the defense [Shejwalkar and Houmansadr, 2021, Bhagoji et al., 2019], as do distributed and durable backdoors [Xie et al., 2020, Wang et al., 2020, Zhang et al., 2022]. C1 quantifies over a *committed* attack, so this literature bounds what any inherited-suppression criterion (ours included) can promise; our single whitebox stress test (§6) measures that gap rather than defending the criterion.

Evaluation methodology, which is where this paper sits. The adjacent precedent is adversarial robustness, where reported numbers proved unreliable until evaluations were required to be adaptive [Croce et al., 2021]. Our point is methodological in the same way but about **identification** rather than attack strength: composition studies gate candidates on standalone effectiveness [Kairouz et al., 2021] and then read the composition’s outcome as evidence about the mechanism, which Proposition 4 shows cannot work at *any* sample size. What is new is the level hierarchy at which the preservation intuition fails and the design class that can test it (Appendix J.4).

The attack side of the same literature engineers *persistence* rather than magnitude: model-scaling [Bagdasaryan et al., 2020] and Neurotoxin [Zhang et al., 2022] both aim at triggers that survive rounds of clean aggregation, which is the quantity we measure as γ (§1) and the reason temporal defense switching does not help in our regime (Appendix O). Nothing in this paper proposes a new aggregator, a new attack, or a new defense; the object of study is the *composition* of defenses that already exist, and the contribution is an identification argument about how such compositions can and cannot be evaluated.

G Notation

symbol	meaning
c_i	upstream coefficient applied to client i ’s update, $T(u_i) = c_i u_i$, $c_i > 0$
A	the set of adversarial participants in a round
ρ	statistic-level dispersion, $\max_i c_i / \min_i c_i$: <i>how far apart</i> the coefficients are
Δ_c	adversary-correlated dispersion, $\mathbb{E}[c_i \mid i \in A] - \mathbb{E}[c_i \mid i \notin A]$: <i>who</i> receives them
r	adversary-to-benign weight ratio, $w_a / \max_b w_b$; the Mode-A dial, $r = e^\nu$
S	the attack’s coordinate extremeness (Theorem 1(1))
m	RFA separation margin, $\Delta'_{\text{sep}}/2(R'_B + \delta')$
Λ_a	aggregate adversarial influence: adversarial share of the Weiszfeld mass
κ	the Mode-S / ladder dispersion dial, $\rho = e^{2\kappa}$
γ	attack <i>persistence</i> , $\text{ASR}(t+1)/\text{ASR}(t)$ after a clean round (≈ 0.66); never a weight

546 H Deployment guidance in full

547 **Deployment guidance.** (0) Check C0: if one candidate defense already suppresses every attack
 548 you have, deploy it alone: composition has nothing to add. (1) Check C1: does at least one defense
 549 suppress known attacks? (2) Check C2 by invariance class (Proposition 1): a cosine-similarity down-
 550 stream defense has its signal preserved behind any positive rescaling, a consensus-distance one does
 551 not. (3) For weighting→aggregator compositions Theorem 1 gives verifiable sufficient conditions
 552 for C2∧C3. Gate on C2; read C3 as a *quantitative diagnostic*: a 5–10 round pilot returns ρ and the
 553 separation margin m , and it is $S > \rho$ (rank aggregators) or $m \gg 1$ (RFA) that matters, not whether
 554 d_1 happens to be heterogeneous, the binary reading that produced both of our false negatives. (4)
 555 Best candidate on our evidence, for further evaluation rather than deployment: **FG→CM** (accuracy
 556 0.68, max ASR 0.107). FG→RFA is marginally lower on the point estimate (0.093 at $n=30$) but
 557 statistically indistinguishable, and carries the *highest* adaptive ASR (0.167) and an 18-point accu-
 558 racy cost, so it is not the more secure choice on our evidence. (5) In our persistent-backdoor setting
 559 fixed-order composition substantially outperforms randomized ordering; we have not tested regimes
 560 where randomization might pay.

561 I Proofs

562 **What the two arms jointly establish.** Both halves are witnessed by runs already reported, so the
 563 following formalizes measurements rather than adding a claim.

564 **Proposition 2** (Statistic-level non-identifiability). *Within the positive per-client rescaling class there*
 565 *exist transform families T_1, T_2 and downstream defenses d_2 such that (a) T_1 and T_2 induce arbitrar-*
 566 *ily different statistics and decisions at d_2 while inducing the same admitted adversarial mass and the*
 567 *same suppression; and (b) T_1 and T_2 induce the same admitted adversarial mass (indeed the same*
 568 *statistic and the same decision) while inducing different suppression. Consequently no predicate of*
 569 *the downstream statistic alone determines the admitted adversarial mass, and no predicate of the*
 570 *admitted mass alone determines suppression.*

571 Both parts are proved by exhibited witnesses (Appendix I): Mode S into Krum under model-scaling
 572 for (a), `cos_krum` for (b). **The corollary is the paper’s title claim:** no metric defined *solely* on
 573 preservation of the downstream statistic can identify security equivalence, (a) exhibiting statistic
 574 disturbance without security change and (b) the converse. **Scope:** a non-existence claim *within the*
 575 *positive-rescaling class on these defenses and attacks*, by exhibited counterexample: **not** that no
 576 metric of any kind can order security.

577 **Proposition 3** (ρ and Δ_c are independently controllable). *Within the same class, the statistic-*
 578 *level dispersion $\rho = \max_i c_i / \min_i c_i$ and the adversary-correlated dispersion $\Delta_c = \mathbb{E}[c_i \mid i \in$
 579 $A] - \mathbb{E}[c_i \mid i \notin A]$ are not functionally related: there is a family with $\Delta_c \equiv 0$ exactly while ρ
 580 ranges over $[1, 54.6]$, and a family with the statistic’s dispersion held fixed while Δ_c ranges over
 581 $[-1.138, +2.469]$. Hence neither quantity determines the other.*

582 This makes §2’s warning precise: **a transform reported by ρ alone leaves Δ_c entirely free.** It does
 583 not follow that Δ_c is a predictor; its magnitude fails to order suppression either (§5). The instrument
 584 is elaborate because §3 leaves nothing simpler. §5 answers what remains: *what happened when we*
 585 *tried it the obvious way first?*

586 I.1 Formal Guarantee: Weighting → Aggregator Class

587 For the practically important class containing our certified compositions (a per-client *weighting*
 588 defense d_1 followed by a *rank-based* or *geometric* aggregator d_2) C2 and C3 can be formally estab-
 589 lished. **Read what follows as a *boundary* theorem and deliberately not a security theorem:** it
 590 says exactly how much an invariance argument about composition can prove, which is preservation
 591 of d_2 ’s discriminative mechanism at levels (P1)–(P3). It bounds no ASR, and the counterexamples
 592 in the box above are counterexamples to reading it as though it did.

593 **Theorem 1** (Bounded reweighting preserves the downstream discriminative mechanism). *Let d_1*
 594 *apply per-client scalar weights $w_i \geq 0$ with ratio $\rho := \max_i w_i / \min_{i:w_i>0} w_i$ (over strictly positive*
 595 *weights), operating on n_b benign and n_a adversarial updates in a round of size K . Assume $w_a > 0$*
 596 *for the adversarial clients; the $w_a = 0$ case is Lemma 1. Then:*

- 597 1. **Rank aggregators.** The composition passes $\{w_i u_i\}$ to an unweighted order statistic over
 598 K transformed values per coordinate. The mechanism is preserved when:
- 599 (i) Count majority: $n_a < K/2$.
 600 (ii) Signed ordering (C2): on suppression-relevant coordinate k , the adversarial trans-
 601 formed value lies strictly beyond every benign one: $w_a u_{a,k} > \max_b w_b u_{b,k}$ (upper
 602 tail) or $w_a u_{a,k} < \min_b w_b u_{b,k}$ (lower tail).
- 603 Sufficient magnitude condition for (ii): if $\text{sign}(u_{a,k})$ is common to all adversaries and
 604 $|u_{a,k}| > \rho \cdot \max_b |u_{b,k}|$, then (ii) holds. This branch is one of two mechanism-preserving
 605 regimes, the other (strong attenuation, $w_a \rightarrow 0$) being Lemma 1.
- 606 2. **Geometric median (RFA):** the mechanism is preserved when separation dominates dis-
 607 placement: the minimum adversarial–benign separation among the transformed updates
 608 strictly exceeds twice the benign spread plus the displacement the adversaries induce in
 609 the geometric median (Appendix I defines the four quantities and proves it), under which
 610 every adversarial residual exceeds every benign one, so RFA assigns adversarial updates
 611 strictly lower inverse-residual weight.

612 *Proofs are in Appendix I. Both parts are sufficient conditions for d_2 to preserve the downstream*
 613 *suppression ordering, not to exclude adversarial contributions outright; **neither is sufficient for low***
 614 *ASR, and part (2) constrains the aggregate adversarial coefficient only through Corollary 1.*

615 Theorem 1(2) orders residuals *per client*, which alone does not bound the adversary’s *total* share
 616 of the aggregate. A realized *separation margin* $m \geq 1$, the factor by which that condition is met,
 617 defined with the bound in Corollary 1 (Appendix I), does bound it, but only quantitatively: as $m \rightarrow 1$
 618 the bound degrades to the adversarial count fraction, so **the separation condition on its own is not**
 619 **enough**. This is the one place the theory predicts a number and the measurement lands inside it: at
 620 the measured $m \geq 2.82$ in every round with $w_a > 0$ (mean 4.71 over those rounds) the bound is
 621 $\Lambda_a \leq 12.6\%$, against a realized $\Lambda_a \leq 1.8\%$. The $m \rightarrow 1$ boundary is where §3’s heterogeneity
 622 failure sits, and it is reached ($m=1.01$ at $\alpha=0.1$).

623 Λ_a is level (P4) in measurable form. (P4) is the first level whose content is the adversary’s
 624 influence on the aggregate, and for a geometric aggregator Λ_a is that level made measurable:
 625 Corollary 1 bounds it, Lemma 1 drives it to a benign-only convex combination, and §4’s instru-
 626 ments intervene on it directly. **So the level the evidence points to is one the theory can bound,**
 627 **just not from a (d_1, d_2) definition pair alone**, which is what a screen would need. Unqualified,
 628 “admission” means *admitted adversarial mass*, whether an adversarial update enters d_2 ’s aggregate
 629 at all; the three aggregator-specific quantities Definition 1 sets out are never used interchangeably
 630 (Appendix G).

631 The remaining case dominates in practice: FoolsGold drives the adversarial weight to zero. This
 632 does *not* remove the adversary from RFA’s input (Definition 2): RFA receives a zero vector, a point
 633 of the geometric-median problem like any other. The correct statement is stronger than removal:

634 **Lemma 1 (Zero-weight annihilation).** *Our implementation runs the ε -smoothed Weiszfeld iteration,*
 635 *whose fixed point $\mu' = \sum_i \lambda_i u'_i$, $\lambda_i \propto 1/\max(\|u'_i - \mu'\|, \varepsilon)$, is a convex combination for every*
 636 *configuration, so no non-degeneracy assumption is needed (Appendix I). If $w_a = 0$ for every ad-*
 637 *versarial client then $u'_a = 0$ for each, and writing $\Lambda_a := \sum_{a \text{ adv}} \lambda_a$ the adversarial terms vanish*
 638 *identically:*

$$\mu' = \sum_{b \text{ benign}} \lambda_b u'_b = (1 - \Lambda_a) \bar{u}, \quad \bar{u} := \sum_b \frac{\lambda_b}{1 - \Lambda_a} u'_b,$$

639 *a convex combination of benign transformed updates. The aggregate is a **contraction of a benign-***
 640 ***only weighted average:** u_a does not appear in μ' at all, and the origin point’s only effect is a smaller*
 641 *effective step: **a utility cost, not a security failure.***

642 The same argument covers CoordMedian. The two cases are exhaustive over every measured round
 643 (FG zeroes all adversaries in 32% of adversary-present rounds, the separation condition holding in
 644 the other 68%), and the mechanism survives for *opposite* reasons: under Lemma 1 the adversaries
 645 hold most of the Weiszfeld mass ($\Lambda_a = 0.71$) but carry no payload; under Corollary 1 they carry
 646 payload but at most 12.6% of it. The contraction $1 - \Lambda_a = 0.29$ is composition’s accuracy cost:
 647 FG→RFA reaches accuracy 0.50 and FG→CM 0.66 against 0.78 for the aggregators alone, paid
 648 even when the zeroed clients are benign.

649 The proofs below are short; they are collected here rather than inline because what each result *rules*
650 *in or out* is the content, not the derivation. Both witnesses of Proposition 2 are runs reported in §4
651 and §5: Mode S into Krum for part (a), `cos_krum` for part (b).

652 *Proof of Proposition 4.* The load-bearing condition excludes d_1 from attaining the minimum in (i),
653 so certification forces $M(d_2, a) < \theta$; both cells with $M(d_2, a) \geq \theta$ are unoccupied whatever the
654 value of P . A contrast varying d_2 therefore compares pairs differing in P only inside the selected
655 region $M(d_2, a) < \theta$, where P and $M(d_2, a)$ are both functions of the same choice of d_2 ; the
656 missing cell is the one in which they vary independently, and no reweighting of the occupied cells
657 recovers it. The emptiness follows from the gate rather than from $\mathcal{D}$. $\square$

658 *Proof of Lemma 2.* C1 at a^* gives $\min_{d \in \{d_1, d_2\}} \text{ASR}(d, a^*) < 0.5$. By C0 the minimum is not
659 attained at d_1 , so it is attained at d_2 . $\square$

660 *Proof of Proposition 5.* Emergence is the negation of the gate, so each member of $\mathcal{E}$ is rejected while
661 belonging to $\mathcal{S}$: a false negative. The bound is the resulting recall. $\square$

662 *Proof of Proposition 2.* Both halves are exhibited. (a) Mode S into Krum under model-scaling, $\kappa=0$
663 against $\kappa=2$: ρ moves from 1 to 54.6 and Krum’s decision changes in up to 0.80 of rounds, while
664 the support of the admitted adversarial mass changes in none of the 60 measured rounds (a floor
665 on this arm: no adversary is admitted at any rung), the adversarial coefficient share is constant at
666 0.266667, and suppression shows **no evidence of a practically meaningful change under the pre-**
667 **specified ± 0.15 equivalence margin** ($|\Delta|=0.026$). (b) `cos_krum` under the pixel backdoor: by
668 Proposition 1(a) its statistic is invariant, and its decision and admission are measured bit-identical
669 (0.000) at every rung, yet ASR falls 0.173 by the first rung (§5). $\square$

670 *Proof of Proposition 3.* By construction: Mode S pins every adversarial coefficient at $c=1$ and
671 spreads only benign clients, giving $\Delta_c \equiv 0$ at every ρ ; Mode A holds the benign weights uni-
672 form and sweeps only the adversary-to-benign ratio, moving Δ_c at fixed dispersion. Realized values
673 are tabulated in Appendix R, computed with no ASR involved. $\square$

674 *Proof of Proposition 1.* (a) $c_i u_i / \|c_i u_i\| = u_i / \|u_i\|$ for $c_i > 0$, so the matrix of pairwise cosines is
675 unchanged; FoolsGold’s weights are a function of that matrix alone (via each client’s normalized
676 maximum similarity), hence unchanged. (c) With $u_1=(1, 0)$, $u_2=(0, 1)$, $u_3=(3, 3)$ the coordinate-
677 wise median is $(1, 1)$ and u_3 is farthest; under $c = (1, 1, 0.1)$ the median becomes $(0.3, 0.3)$ and u_3
678 becomes the *closest* point, reversing the ordering Reputation depends on. $\square$

679 *Proof of Theorem 1.* (1) CoordMedian operates on $\{w_i u_{i,k}\}_{i=1}^K$ per coordinate. If the adversar-
680 ial values lie strictly beyond all benign values on one side (ii) and $n_a < K/2$, then at most
681 $n_a < \lceil K/2 \rceil$ of the extreme order statistics are adversarial, so the median is a benign value. For
682 the magnitude condition: $w_a > 0$ gives $\rho \geq \max_b w_b / w_a$, so $|u_{a,k}| > \rho \max_b |u_{b,k}|$ implies
683 $|w_a u_{a,k}| > w_a \rho \max_b |u_{b,k}| \geq (\max_b w_b) \max_b |u_{b,k}| \geq \max_b |w_b u_{b,k}|$; with a common sign this
684 places $w_a u_{a,k}$ strictly beyond every benign value in that direction. Model-scaling sets $u_a = S \tilde{u}_a$
685 with $S=10$, which amplifies the adversary’s own direction and so satisfies the sign assumption by
686 construction; with measured $\rho \leq 2.1$ the margin is $S/\rho \geq 4.8$. TrimmedMean additionally needs
687 the trimming budget to cover the adversarial count.

688 (2) Every benign residual satisfies $\|u'_b - \mu'\| \leq R'_B + \delta'$. For adversarial u'_a : $\|u'_a - \mu'_B\| \geq \Delta'_{\text{sep}} - R'_B$,
689 so $\|u'_a - \mu'\| \geq \Delta'_{\text{sep}} - R'_B - \delta'$. Requiring this to exceed $R'_B + \delta'$ gives (i). $\square$

690 Theorem 1(2) orders residuals *per client*, which alone does not bound the adversary’s *total* share of
691 the aggregate: several adversaries could each be individually down-weighted yet collectively hold
692 substantial mass. The separation margin controls this, quantitatively:

693 **Corollary 1** (Bounded aggregate adversarial influence). *Let $m := \Delta'_{\text{sep}} / [2(R'_B + \delta')]$ be the realized*
694 *separation margin, so Theorem 1(2) reads $m > 1$. Benign residuals are at most $D_b := R'_B + \delta'$ and*
695 *adversarial residuals at least $(2m - 1)D_b$, so bounding $\sum_j 1/d_j$ below by the benign terms alone*
696 *gives*

$$\Lambda_a = \sum_a \lambda_a \leq \frac{n_a}{n_b(2m-1) + n_a}.$$

697 *The aggregate is thus a convex combination whose adversarial share is at most Λ_a : writing $\mu' =$*
698 *$(1 - \Lambda_a)\bar{u}_B + \Lambda_a\bar{u}_A$ for the benign and adversarial parts, the displacement from the benign-only*
699 *combination $\bar{u}_B$ is at most $\Lambda_a\|\bar{u}_A - \bar{u}_B\|$.*

700 The bound is only as good as the margin: as $m \rightarrow 1$ it degrades to $n_a/(n_b + n_a)$, approaching
701 40% of the mass at $n_a=2, n_b=3$. We measure $m \geq 2.82$ in every round with $w_a > 0$ (mean 4.71
702 over those rounds), capping the adversarial share at 12.6% for $n_a=2$; the realized Λ_a never exceeds
703 1.8%, and the bound holds in all 17 measured rounds.

704 J Body material, at full length

705 The body compresses the following; nothing here is a new claim, a new number or a new run.
706 Figure 2 summarizes the preservation chain and the two directions in which it breaks.

707 J.1 Contributions, organization and scope in full

708 §1 states these compressed. Nothing here is a new claim.

709 J.1.1 The three evidence tiers

Table 2: The three tiers of evidence, of very unequal weight and never pooled. §1 states them in prose; every empirical claim in the paper is tagged with one of the three words where it is made.

tier	design	what it licenses
Causal	within-defense intervention at fixed d_2 , attack and C1 input: Mode S, Mode A (§4)	statements about the <i>mechanism</i> , scoped to these cells
Replication	EMNIST-byclass, second dataset <i>and</i> architecture ($\Delta = -0.017$), and <code>coord_median</code> , a second invariance class (+0.098)	“replicated on a second dataset and architecture”, never “generalizes”
Screening	the outcome-gated 42-pair suite, 38/42	evaluation prioritization only ; by Proposition 4 it cannot identify the mechanism’s contribution at <i>any</i> sample size

710 J.1.2 The five contributions

- 711 1. **The preservation hierarchy, and the two-directional break above.** Because the per-client
712 defenses in use are all positive scalar rescalings, mechanism preservation is a decidable *invari-*
713 *ance* question (Proposition 1), and on the weighting→aggregator class we prove it with explicit
714 margins (Theorem 1), a **boundary** theorem, not a security theorem. Definition 1 gives the five
715 levels the box states, and §4 measures both breaks. One corollary is worth stating alone: weight
716 heterogeneity is safe at *both* extremes and dangerous in the middle, so **the weight ratio ρ alone**
717 **mispredicts** (Lemma 1, §R.1).
- 718 2. **An identification boundary for outcome-gated screening, and the design class it leaves**
719 **admissible.** Proposition 4 proves the general claim above; its FL instance is that $C0 \wedge C1$
720 force the downstream defense of every certified pair to suppress the attack alone, **mechanism-**
721 **effectiveness confounding** (§3), and its evidential cost is real: a matched ablation built to break
722 the confound had its premise refuted by its own pre-freeze baselines, so **the confound re-**
723 **materialized inside the experiment designed to remove it.** A second boundary limits coverage
724 rather than validation: **emergent** suppression (present in the pair, absent in both constituents)
725 cannot satisfy an inherited-suppression gate, capping such a screen’s recall (Proposition 5), and
726 our own best composition is its witness.
- 727 3. **Three central empirical findings, all causal interventions, not a validation of any screen.** (i)
728 *The negative, instrumented and replicated* (Causal, then Replication). Mode S pins every ad-
729 versary’s coefficient, closing Lemma 1’s attenuation channel *by construction*; a **pre-registered**
730 **replication** in a *different* invariance class and on a *second dataset and architecture* reproduces it

while **refuting the cross-arm admission ordering we had predicted** ($R^2=0.239$ over 26 cells), so **admission is a direction to measure, not a quantitative predictor**. (ii) *A sign reversal under a change of intervention (Causal)*. The same cell yields -0.272 ($p_{\downarrow}=0.0015$) with adversarial influence free to co-vary and $+0.098$ with it pinned: the confounded design does not lose power, **it gets the sign wrong**; hence we withdraw the pooled form of C2’s causal claim (80 runs; $\rho_s = +0.351$, $p = 0.091$; §5). (iii) *Heterogeneity closes the theorem’s own margin*. At Dirichlet $\alpha=0.1$ RFA’s all-round mean separation margin collapses from $m=3.24$ to $m=1.01$, the limit at which Corollary 1’s bound goes vacuous, and two of three certified compositions cross the threshold (§3). A fourth predictor falls with them: **adversary-correlated dispersion** Δ_c is independently controllable from ρ (Proposition 3) and locates the two real anchors our synthetic curve misses, but **orders suppression no better**. Statistic disturbance, admission magnitude and coefficient targeting are each refuted *as a predictor*: what survives is a *preservation level*, not a statistic of the coefficient vector.

4. **The screening rule is a consequence of (2), not a claim of its own** (*Screening* throughout). What survives the negative is a conservative *evaluation-prioritization* rule: read C2 off the downstream signal’s invariance class, then evaluate the survivors first. We report it as an application (§3), never as a validated security predictor; it certifies only *inherited* robustness under a *committed* attack, at recall 40%, and outputs an *evaluation ordering*, never a deployment decision.

5. **Secondary results**. Fixed-order composition beats randomized ordering $9\times$, because preservation conditions are order-sensitive; and **one** whitebox composition-aware *adaptive* stress test raises $FG \rightarrow RFA$ to 0.167 ± 0.069 , $3.3\times$ its seed-matched committed 0.051, **an inherited-robustness guarantee is not adaptive robustness** (§6, Appendix O).

J.1.3 How the paper is organized, as an argument

Composition creates an **identification problem**. §2 defines the object to be identified: mechanism preservation as invariance, the five levels, the conditions; §3 characterizes the **boundary** on identifying it and *derives* the class of designs that can; §4 runs the experiment that class dictates; §5 reports what the boundary **falsifies**, our own pooled hypothesis included; §3 states what survives.

J.1.4 Why this class, and the scope discipline

Everything we *prove* is instantiated for positive per-client rescalings, deliberately: the class is a **laboratory for studying composition, not a general model of it**, it covers the per-client defenses in current use (norm clipping, FoolsGold, RFA reweighting, FLTrust) and makes invariance decidable from definitions. **Every claim here is general, formal or empirical, and they do not transfer upward**. Table 1, first in the appendix, is the one place each scope condition is stated (what falls outside the class, what the theorems do and do not bound, the sample sizes, what “refuted” means, why our instruments are not deployable) and we do not restate them at each use.

J.1.5 Threat model and scope

We consider persistent backdoor attacks where an adversary controlling fraction f of clients embeds triggers that survive across rounds, and define *persistence* γ as the fraction of backdoor ASR retained per round of clean aggregation ($\gamma \approx 0.66$ for CIFAR-10/CNN). Attacks include model-scaling [Bagdasaryan et al., 2020], pixel-pattern backdoors [Chen et al., 2017], distributed backdoor attacks [Xie et al., 2020], and adaptive composition-aware adversaries [Shejwalkar and Houmansadr, 2021]. The framework constrains the aggregation pipeline, not the communication graph, and is evaluated only in **star-topology FL**. **A scope condition on the criterion itself, not a limitation of our runs**: C1 and every prediction built on it are defined against a *committed* adversary, so what is bounded is *inherited* robustness under a fixed attack, **never adaptive robustness** (Table 1).

J.2 The prioritization criterion, and where it breaks

This section is a consequence of the boundary, and what survives is an evaluation-prioritization criterion, not a security screen. $C1 \wedge C2$ answers one question: given a menu, which ordered compositions are worth running first. **We disclaim predictive validity explicitly**; it does not predict whether a composition is secure, its positive direction is unvalidated, and the cross-arm admission ordering a predictive reading would need was *refuted* by our own pre-registered replication (§4). It

also costs us our own best composition: under the per-attack quantifier **FG→RFA no longer satisfies C1**, so the lowest-ASR composition we found is a **false negative** and Proposition 5’s witness (uncertifiable by *any* inherited-suppression screen rather than missed by ours) while the certified in-distribution set (FG→CM 0.107, Rep→CM 0.423) is correct and the miss costs nothing detectable (0.093 against 0.107, Welch $p=0.52$). Out of sample it licenses *positive* predictions only, against uninformative FAIL controls, so **this is not an out-of-sample classification result**: the *Screening*-tier 38/42 at recall 2/5 rests on the in-distribution tiers. The defensible claim is *conservative prioritization*; scoring condition by condition, the confusion matrix, the false negatives and what the rule can and cannot do are in Appendices S–T, U and Table 9 (App. V).

The procedure, and where it breaks. (0) If one defense already suppresses every attack you have, stop (C0); (1) require a constituent that suppresses the committed attack (C1); (2) read C2 off the downstream signal’s *invariance class*, which §3 says you cannot test directly; (3) read C3 as a margin from a short pilot, never as a gate. Then **evaluate the survivors first, and evaluate them**: none of this certifies security. **Where the rule breaks is measured rather than argued: under extreme heterogeneity (Dirichlet $\alpha=0.1$) two of three certified compositions cross the threshold and RFA’s all-round mean separation margin collapses from $m=3.24$ to $m=1.01$** : the $m \rightarrow 1$ limit at which Corollary 1’s bound says nothing. Theorem 1’s guarantee is therefore not merely conservative under heterogeneity but *empty* there, and the theorem earns its keep by predicting its own failure region: a short pilot measuring $m \approx 1$ declines to deploy. **Post-hoc, not pre-registered** (Appendix O.1).

802 J.3 What is proved, what is supported, and the six-step procedure

What is proved, what is supported, and what is not. **Proved**: the invariance classes (Prop. 1); sufficient conditions for mechanism preservation on the weighting→aggregator class (Thm. 1, Lem. 1); the identification boundary and the design class it leaves admissible (Prop. 4, Lem. 2); the recall bound on inherited-suppression screens (Prop. 5). **Empirically supported** (*Causal*, then *Replication*): that substantial disturbance of the downstream statistic need not produce a corresponding change in suppression when the adversarial coefficient share is held fixed, on the Krum arm under Mode S and, by pre-registered replication, in a *different* invariance class and on a *second dataset and architecture*; and that the criterion behaves as a conservative prioritizer on 42 pairs. **Not established**: its positive predictive direction ($R^2=0.239$, cross-arm ordering refuted), and generalization beyond the scope Table 1 fixes, $\gamma \approx 0.66$ being architecture-conditional. The criterion **trades discovery for reliability**: it rejects working compositions (recall 40%, one of them ours), cannot in principle do otherwise (Prop. 5), and draws negative-direction evidence only from the in-distribution tiers. FG→CM is an evaluation *ordering*, **not a deployment recommendation** (Appendix H).

What to do instead: six steps for any composition claim. (1) Name the downstream mechanism and its invariance class. (2) State *which* preservation level you claim, (P1)–(P5). (3) Check the upstream transform lies in the class your argument covers: C0 here; one altering participation, carrying cross-round state or mixing clients is covered by no invariance argument. (4) Hold d_2 , the attack, and therefore the gate’s input fixed; **do not select candidates on the outcome**. (5) Intervene on the upstream transform and **measure the channel**: aggregate displacement, decision, admission, influence, not the outcome-conditional contrast. (6) Report the level at which the chain broke. Steps (4)–(5) are what Proposition 4 licenses, and following them turned our -0.272 into $+0.098$; **this, not the criterion, is what we ask readers to carry away**. One boundary it does not cross: under *one* whitebox composition-aware *adaptive* stress test FG→RFA rises to $ASR = 0.167 \pm 0.069$, $3.3\times$ its seed-matched committed 0.051, so **an inherited-robustness guarantee is not adaptive robustness**: the condition the framework most lacks (Appendix O).

828 J.4 The identification boundary, formally

This section states §3’s two propositions and their FL instance at full length; the body gives the boundary in words and the design it dictates.

Proposition 4 (Why constituent-effect contrasts cannot identify composition effects under inherited-effect gating). *Let a screen certify $(d_1 \rightarrow d_2)$ against attack a only if both (i) an **eligibility gate** $E(d_1, d_2, a) := \min_{d \in \{d_1, d_2\}} M(d, a) < \theta$ holds, where M and θ are the same outcome metric*

834 and threshold the screen predicts for the composition, and (ii) a **mechanism condition** $P(d_1, d_2)$
835 holds, determined by the two defenses' definitions alone. Suppose the screen is applied only where
836 the downstream mechanism is load-bearing, $M(d_1, a) \geq \theta$. Consider estimating P 's contribution
837 to $M(d_1 \rightarrow d_2, a)$ from contrasts across a defense menu $\mathcal{D}$. Then every certified pair satisfies
838 $M(d_2, a) < \theta$, so of the four cells of $\{P, \neg P\} \times \{M(d_2, a) < \theta, M(d_2, a) \geq \theta\}$ the certified set
839 occupies at most two, and **the cell needed to separate P from d_2 's standalone effectiveness is empty**
840 **by the gate, for every $\mathcal{D}$. No contrast varying d_2 identifies P 's contribution. Identification requires**
841 **holding d_2 , a and therefore E fixed while intervening on the upstream transform: the admissible**
842 **designs are within-defense.**

843 The structure is a collider: the gate E is a function of constituent effectiveness, effectiveness affects
844 the composition's outcome, and the mechanism condition P is fixed by the same pair of definitions,
845 so conditioning on E makes P and effectiveness co-vary across every certified candidate. The design
846 table is the whole argument, and **the missing counterfactual is a cell, not a sample size:**

	$M(d_2, a) < \theta$ (d_2 suppresses a alone)	$M(d_2, a) \geq \theta$ (it does not)
847 P holds (mechanism preserved)	observed: every certified pair	empty by the gate
$\neg P$ (mechanism destroyed)	observed: predicted-FAIL pairs	empty by the gate

848 Reading down a column contrasts P against $\neg P$ at fixed effectiveness (the comparison that would
849 identify P 's contribution) and the right-hand column, the only one in which d_2 is not already doing
850 the work, is empty for every menu $\mathcal{D}$. **The counterfactual needed is excluded by design, not**
851 **undersampled.** The emptiness follows from the gate, not the menu, and nothing in the argument is
852 specific to FL, to backdoors, or to our conditions (Appendix I). Its instance for *our* criterion is one
853 line:

854 **Lemma 2** (C0 and C1 place the effective constituent downstream). *Let $(d_1 \rightarrow d_2)$ satisfy C0 and C1,*
855 *and let a^* be a committed attack witnessing C0, so $\text{ASR}(d_1, a^*) \geq 0.5$. Then $\text{ASR}(d_2, a^*) < 0.5$:*
856 *the downstream defense suppresses a^* on its own.*

857 Instantiated: in *every* in-scope pair our conditions certify, d_2 already suppresses the attack d_1 fails,
858 so a certified composition's low ASR is always attributable to d_2 's standalone strength (C1) rather
859 than to d_1 having preserved d_2 's mechanism (C2, C3): confounded **by the conditions themselves**,
860 not by an unlucky menu. We name this **mechanism–effectiveness confounding**.

861 **This is selection on the outcome; where it hides is the point.** The hazard is standard (condition-
862 ing on a descendant of both the putative cause and the outcome [Elwert and Winship, 2014, Pearl,
863 2009]) and **we claim no novelty for it**. What is not standard is that in composition studies the gate
864 is presented not as conditioning but as a *reasonableness* filter (“we study defenses that work”), so it
865 reads as hygiene rather than as the step that removes identification. Two consequences are routinely
866 assumed away: **a better estimator cannot fix it**, no reweighting of occupied cells recovering an
867 unoccupied one, and **the gate cannot be dropped**, because without it C1 is vacuous.

868 **Both worked instances are ours, and one of them inverted a sign.** We name no other paper;
869 our own two designs are the instances. The C1-gated study of §5 is the template at the level of the
870 *criterion*; the dose ladder of §5 is a second instance one level down, its gated quantity being the
871 dose's incidental attenuation of whichever client carries the poison. **We ran the outcome-gated**
872 **design first, and on the identical cell it returned the opposite sign** (Table 3): the hazard cost us
873 the sign of our own effect, in the direction that flattered us.

874 **The admissible design, derived: this, not the criterion, is the practical contribution.** Read
875 forwards the boundary is *constructive*: each clause below is forced by a term in Proposition 4, not
876 chosen for convenience. (1) **Hold d_2 and the attack fixed**, so the gate's input $M(d_2, a)$ is numeri-
877 cally the same quantity on both sides of every contrast and cannot co-vary with the mechanism. (2)
878 **Intervene on the upstream transform** instead of selecting a different d_2 off a menu, because it is
879 varying d_2 that empties the identifying cell. (3) **Pin the adversarial coefficient channel**, because
880 an upstream that attenuates the adversary moves the outcome by a route with nothing to do with d_2 's
881 statistic: the confound that inverted our own sign above. (4) **Measure every channel: aggregate**
882 **displacement, decision, admission, influence, rather than the outcome-conditional contrast, the**

outcome being exactly the variable the gate conditioned on. §4 is clause (3) built as an instrument; Appendix J.3 states all four as a six-step procedure for any composition claim.

A second boundary: what such a screen cannot find. The gate that limits what a screen can be *validated* on also limits what it can *cover*, which converts the most obvious objection to our criterion, that its recall is poor, into a scope result.

Proposition 5 (Inherited-suppression screens cannot certify emergent synergy). *Call $(d_1 \rightarrow d_2)$'s suppression of a **emergent** if $M(d_1 \rightarrow d_2, a) < \theta$ while $\min_{d \in \{d_1, d_2\}} M(d, a) \geq \theta$. No emergent composition satisfies the eligibility gate of Proposition 4. Hence, writing $\mathcal{S}$ for the compositions with $M < \theta$ and $\mathcal{E} \subseteq \mathcal{S}$ for the emergent ones, any such screen's recall obeys $\text{recall} \leq 1 - |\mathcal{E}|/|\mathcal{S}|$ on every menu and every attack.*

This is a scope theorem, not a defect found by tuning, and it is specific neither to C1 nor to FL: any screen requiring a constituent to suppress the attack independently inherits it. Our own lowest-ASR composition is the witness (FG→RFA reaches 0.093 where FoolsGold alone reaches 0.732 and RFA alone 0.845), so the bound binds at $|\mathcal{E}|/|\mathcal{S}| = 1/5$ (§3). Screening for inherited robustness and searching for synergistic compositions are different problems; only the first is ours.

What the boundary rules out, our own previously-decisive test included. These proofs are short and their value is what they rule in and out. They rule *out* the experiment we had previously named as decisive: a d_2 whose signal is invariant under positive rescaling but *weak in isolation* cannot satisfy C1 in scope, so no such certified pair exists to test. §4–§5 run four members of the class they rule in. **The strongest disturbs the statistic while pinning the adversarial mass, isolating statistic disturbance from adversarial coefficient attenuation, so we built it (§4).**

J.5 The targeted suite in full

The body compresses these paragraphs; nothing here is a new claim or a new run.

Reputation's confirmation, in full. With the adversary's share pinned, reputation's rise at $\rho=54.6$ shrinks from +0.762 to +0.178 (one-sided Welch $t = -5.22$, $p=0.0017$); what remains tracks a small *admission* change (0.000 → 0.020) rather than the large decision change (0.000 → 1.000).

The arc that produced this section. Our pre-registered pooled hypothesis (H1) (ASR should rise with measured disturbance of the statistic the downstream defense reads) was **rejected**, the effect coming out *negative* (-0.272 , $p_{\downarrow}=0.0015$ on the flagship cell), because that ladder moved adversarial influence and the statistic *together* and so never tested H1 at all (§5). This section builds the instrument that fixes it, pinning the adversarial coefficient share while the statistic moves, so statistic disturbance is isolated from adversarial coefficient attenuation. C2 is written at level (P2); if losing (P3) meant losing (P5) the criterion would transfer straight to security, and the experiment below adjudicates that implication by **oracle intervention on two arms in two invariance classes** (*Causal* tier), not by a law about defense compositions in general.

An instrument that closes the attenuation channel by construction. A positive rescaling reaches d_2 through two channels at once: it disturbs the *statistic*, and it *attenuates* whichever client carries the poison (Lemma 1). Separating them needs an upstream that moves the first while pinning the second. **Mode S** does this: it fixes every adversary's coefficient at $c=1.0$ exactly and spreads only the benign clients over $\rho = e^{2\kappa}$, so the **adversarial coefficient share is constant at 0.266667 across all four rungs** (spread 3.5×10^{-7}), and a **rise in ASR here cannot be attenuation**. Design and thresholds were frozen before any run, the share verified with no ASR involved (Appendix X). **Mode A** sweeps the payload instead (§5). **Both are oracle instruments, not defenses and not models:** they read adversary identity, which no deployable defense can, and they do **not** quantitatively reproduce any real defense's behaviour, at matched disturbance two real upstream anchors sit +0.767 and +0.789 above the synthetic curve (§5), the measured size of that gap. Their purpose is causal identification, not defense evaluation.

Two readings of C2, frozen against each other. *H-statistic:* what matters is disturbance of d_2 's statistic, read off its *decisions*. *H-admission:* what matters is the *adversarial input* d_2 *admits*. On the Krum arm these predict opposite outcomes, since Mode S disturbs Krum's selection heavily while, the adversary's weight being pinned, leaving the adversarial mass it admits untouched. That

arm adjudicates (Table 4, App. N): as ρ climbs to 54.6, Krum’s **decision changes in up to 0.80 of rounds** while its **admission changes in none of the 60 measured rounds**, and ASR shows **no evidence of the predicted increase, remaining inside the pre-specified equivalence margin** ($p_{\uparrow}=0.70$; $|\Delta|=0.026 < 0.15$). **H-statistic is refuted**: on this arm **substantial statistic and decision disturbance does not produce a corresponding change in suppression once the adversarial coefficient share is held fixed**, and the evidence points to admission as the relevant channel in these experiments.

What the instrument does *not* hold fixed, named in full, and what it pins exactly. Mode S pins the *adversarial* coefficients; it does not pin the benign ones and was never claimed to. So the dose also moves (a) Krum’s pairwise distances, hence the statistic itself; (b) *which* benign client Krum selects, in 0.533/0.800/0.733 of rounds at $\rho=2.72/7.39/54.6$; (c) the **magnitude and geometry of the update actually aggregated**, since the selected client enters scaled by its own coefficient; and (d) the model trajectory, which diverges across rungs in consequence. Channel (c) is **already measured and printed**: the Δ agg. column of Appendix C is $\|\text{agg}(T(U)) - \text{agg}(U)\|/\|\text{agg}(U)\|$, the relative displacement of the aggregate the defense *emits*, and reads **0.892** here. Appendix D splits that 0.892 into 0.986 of *re-selection* and 0.973 of *rescaling*, with none of the re-selections landing on an adversary. So the claim is sharper: **that displacement, with decision changes in up to 80% of rounds, leaves suppression inside the pre-specified equivalence margin, provided the adversarial coefficient share does not move. What is pinned is pinned exactly**: $c_{\text{adv}}=1.0$ at every rung with maximum deviation 0.00e+00 and the share constant to 3.5e−07 on CIFAR-10, 3.9e−07 on EMNIST-byclass, float32 read-back noise, not slack in the instrument (`measure_admission.py`).

Closing the magnitude channel as well: the pre-registered score-only control. Channel (c) is the one confound Mode S leaves open, so we closed it too. **Score-only Krum** scores on the transformed stack exactly as before and aggregates the *untransformed* selected update, so the dose changes *which* client contributes and nothing about *what* it contributes: Δ dec. is the same 0.733 while Δ agg. is no longer 0.892 but the re-selection component alone, 0.986 relative to $\|u_{s_0}\|$. Rules were frozen at 35788d9 before any score-only ASR existed: primary Δ against the same ± 0.15 margin, where $\Delta > +0.15$ would have said **the published result was partly an artifact of the magnitude channel** and narrowed the central claim. **The outcome is $\Delta = -0.023$** (mean ASR 0.062 $\rightarrow$ 0.039) against -0.026 uncontrolled on the identical cell, so **closing the magnitude channel moves the effect by 0.003**; the refuting threshold 0.212 is nowhere approached and every rung clears the accuracy floor. **The central claim is thus instrumented against two independent confounds**, attenuation and magnitude (row six of Appendix C). The secondary trend test, the headroom asymmetry, the selectors-only scope and the `cos_krum` corollary are in Appendix E.

A pre-registered replication, on a different defense class, that refutes the positive half. The two existing Mode-S arms move together on both predictors, so neither adjudicates. `coord_median` under the pixel backdoor is *rank-discordant*: of the interpretable arms it has the **smallest decision change** (0.482) and the **largest admission-level change** ($\Delta\Lambda_a=0.033$), so the two readings predict opposite rank positions for its rise. Thresholds were frozen against the two published rises: confirm if $\Delta > +0.178$, refute if $\Delta < +0.150$ (Appendix X). **The outcome is $\Delta = +0.098$, so the admission ordering is refuted**: the arm with the largest admission-level change rose *less* than one with a smaller change. Nor does the decision reading predict it, having called the *smallest* rise when $+0.098$ is the middle one; the added cells leave the pooled model unconfirmed ($R^2=0.239$, the share coefficient’s sign failing the pre-specified conjunction). **What replicates is the negative**, on a coordinate-wise order statistic rather than Krum’s selector: a decision change of 0.482 costs no suppression by the frozen margin, at accuracy 0.698. Two qualifications: this cell’s identity ASR is 0.443, so equivalence is to the *identity rung*, not a low absolute ASR; and the rise is positive in 5/5 seeds (post-hoc), so the cost is **small and consistent, not zero. The instrument earns its keep**: the confounded ladder on this identical cell *falls* 0.272 ($p_{\downarrow}=0.0015$), so closing the attenuation channel **inverts the sign of the effect**.

What the two arms jointly establish. Both halves are witnessed by runs already reported, so what follows formalizes measurements rather than adding a claim. **Statistic-level non-identifiability** (Proposition 2, proved from the exhibited witnesses in Appendix I): within the positive per-client rescaling class there are transform families inducing arbitrarily different statistics and *decisions* at the *same* admitted adversarial mass and the same suppression (Mode S into Krum under model-

scaling) and families inducing the same statistic, decision *and* admitted mass at *different* suppression: `cos_krum`. Consequently **no predicate of the downstream statistic alone determines the admitted adversarial mass, and no predicate of the admitted mass alone determines suppression. That corollary is the paper’s title claim:** no metric defined *solely* on preservation of the downstream statistic can identify security equivalence. **Scope:** a non-existence claim *within this class on these defenses and attacks*, by exhibited counterexample: **not** that no metric of any kind can order security. A second proposition sharpens §2’s warning: statistic-level dispersion ρ and adversary-correlated dispersion Δ_c are **independently controllable** ($\Delta_c \equiv 0$ while ρ ranges over $[1, 54.6]$; Δ_c over $[-1.138, +2.469]$ at fixed dispersion), so **a transform reported by ρ alone leaves Δ_c entirely free** (Proposition 3), which does *not* make Δ_c a predictor, its magnitude failing to order suppression either (§5). The instrument is elaborate because §3 leaves nothing simpler. §5 answers what remains: *what happened when we tried it the obvious way first?*

1001 K What the boundary licenses: the suites in full

1002 This section holds §5’s three suites at full length: the two attempts the identification boundary
1003 defeated, the confounded dose ladder that motivated Mode S, and the remaining targeted arms.
1004 Nothing here is rescored; the body states the outcomes.

1005 K.1 Two attempts that the confound defeated

1006 Both were built to validate the *screen*; neither could, so they are evidence for Proposition 4 rather
1007 than for the framework.

1008 **Two pre-registered suites, and the two ways the boundary asserted itself.** (i) *A prospective*
1009 *suite on three unused defenses:* 13/20, and *no positive test*. Everything scored in §3 was developed
1010 on the menu it is scored on, so we ran a prospective test on FLTrust [Cao et al., 2021], Krum and
1011 Multi-Krum [Blanchard et al., 2017], for which the theory predicts *opposite* outcomes. All 20 labels
1012 were git-committed before the output directory existed and both attacks clear the FedAvg sanity
1013 gate, so the negatives are informative (11/13 in scope, 5/7 on the falsifiable arm; Appendix Q).
1014 **Two disclosures dominate it:** FLTrust alone suppresses both attacks (0.048), so its 7 pairs are **out**
1015 **of scope** by C0 (applied *retrospectively*) and its 6 in-scope predicted-LOW pairs are **near-identities**,
1016 four bit-identical to FLTrust alone, so **the suite supplies no falsifiable prospective positive test.** (ii)
1017 *A matched metric-swap ablation, whose premise its own baselines refuted.* To break the confound by
1018 construction we held the suppression *principle* fixed and varied only the invariance class, mirroring
1019 Reputation and Krum scoring from *directions* alone. Measuring the C1 input before freezing **refuted**
1020 **the design’s premise:** `cos_reputation` collapses ($0.017 \rightarrow 0.982$) and `cos_krum` reaches 0.078
1021 only at 0.146 accuracy, failing the C1 gate applied everywhere else (**magnitude information is**
1022 **what lets this family handle a magnitude attack**), so C1 bound for all eight pairs, every label
1023 was HIGH, and the suite bears on specificity only (8/8; App. L, P). **The confound re-materialized**
1024 **inside the experiment built to remove it.**

1025 **The one contrast that escapes, and the design it dictates.** One comparison evades the boundary:
1026 hold d_2 and the attack fixed and vary only the upstream, so C1 is identical by construction. On
1027 *Krum/model-scaling*, NormClip disturbs its selection 0/9 and reaches 0.062 while RFA disturbs it
1028 6/9 and reaches 0.779 (Welch $p=0.012$, $n=5$), with no overlap across the six cells where standalone
1029 d_2 genuinely suppresses. **This is post-hoc** (Appendix L) but has what nothing above it does: C1
1030 held exactly fixed, disturbance measured on the mechanism rather than read off the outcome.

1031 K.2 H1, and the confounded design that motivated the instrument

1032 Both suites above tried to validate C2 by *choosing a different downstream defense*: the move Propo-
1033 sition 4 rules out. The admissible alternative varies only the *dose* at fixed d_2 and attack, so the C1
1034 input is the same quantity at every rung and the confound is **impossible by construction** rather than
1035 matched away.

1036 **The instrument, the abscissa, and the predictions frozen against them.** The synthetic d_1 has
1037 one parameter, the *dispersion* of a positive per-client rescaling: $c_j \propto \exp(\kappa(2j/(K-1) - 1))$ at

position j of a (seed, round)-keyed permutation, normalized so $\text{mean}_j(c_j) = 1$. It is Proposition 1's form, the **dose** is $\rho = e^{2\kappa}$, $\kappa=0$ is bit-identical to d_2 standalone, and normalizing the mean while keying the permutation on (seed, round) holds aggregate scale fixed and stops the dose parking adversaries at a fixed end: neither of which, as it turns out, pins Δ_c . Frozen grid $\kappa \in \{0, 0.5, 1.0, 2.0\}$, four cells across three invariance classes. What matters to C2 is not ρ but what the rescaling does to the statistic d_2 reads, so we measured that first, with no ASR involved, establishing that *there is a dose* (0 at identity, rising to 0.87/0.93 for class (c)) and that *class-(a) invariance holds in code*. *Primary, pooled*: across all 16 (arm, κ) cells the ASR rise over that arm's own $\kappa=0$ rung is positively rank-correlated with measured disturbance (Spearman, one-sided, $\alpha=0.05$). *Secondary, per arm*: monotone rise for class (c), **FLAT** (the frozen label, scored as equivalence) for class (a), shallower for class (b); one-sided Jonckheere–Terpstra at $n=5$, “flat” scored as *equivalence* ($|\Delta| < 0.15$, every rung below 0.5), a margin fixed in advance. **The ladder was falsifiable both ways**: `cos_krum`'s aggregate is still $c_j u_j$ for the selected j , so suppression can be lost through the magnitude channel, and a class-(c) arm inside the margin would refute the central fragility claim. **No frozen prediction uses S** : measured, S is **not a stable per-attack constant** (medians 0.57/0.92, range to 12.59), a limit on the theorem's checkability. Design, grid, arms, shapes and refutation criteria were frozen in advance (Appendix M).

Outcome: one arm confirmed, two refuted, one indeterminate; the pooled test fails. All 80 runs completed and were scored per seed against the frozen rules (Table 5, App. N). The **primary** pooled prediction is **refuted**: Spearman $\rho_s = +0.351$ over 16 cells, one-sided $p = 0.091$, against a frozen $\rho_s > 0$ at $p < 0.05$; **in three of four arms ASR falls as the dose rises**. Arm 2 confirms decisively: reputation $0.017 \rightarrow 0.779$ ($p = 1.7 \times 10^{-6}$) at nearly constant accuracy, the paper's only prospective positive for C2, and it establishes the *destruction* half only. Arm 1 refutes more than its own shape: Krum, whose statistic the fragility claim is stated over, shows no rise (inside the equivalence margin) despite measured disturbance 0.867, and does not reproduce our within-defense contrast ($0.062 \rightarrow 0.779$ under RFA at a *lower* disturbance), so selection-flip rate cannot be its mechanism. Arm 3 is indeterminate and uninterpretable anyway (`cos_krum/pixel` falls $0.490 \rightarrow 0.086$, the last rung below the accuracy gate) and arm 4 is refuted outright (`coord_median/pixel` falling $0.443 \rightarrow 0.170$ at accuracy ≥ 0.69). Both fell where a rise was frozen, which is why neither is scored in our favour.

Two faults, both ours, and now measured rather than conjectured. *The instrument.* Normalizing $\text{mean}(c)=1$ fixes the aggregate's scale but not any client's *relative* weight, and because the dose ignores adversary status a $55 \times$ spread mostly *dilutes* whichever client carries the poison: on the same rounds the ladder's Δ_c drifts monotonically negative with the dose ($0.000 \rightarrow -0.140 \rightarrow -0.240 \rightarrow -0.336$), so three arms measure attenuation rather than C2 disturbance. **The adversary-targeted dose is the fix**: Mode S is the Δ_c -**pinned** instrument (Δ_c exactly 0 at every rung, ρ up to 54.6) and Mode A the Δ_c -**swept** one. *The abscissa.* At matched disturbance two of six *real* upstream anchors miss the synthetic curve badly (RFA $\rightarrow$ Krum $+0.767$, NormClip $\rightarrow$ Reputation $+0.789$), and not through a range mismatch; Δ_c locates the discrepancy, as Proposition 3 says it may, those two being the most extremely targeted of the four real transforms ($-1.21, -0.72$) at ρ of only 14.8 and 4.19 against the ladder's 54.6: **matched on the statistic, badly unmatched on the targeting. But this rescues no predictor**: Δ_c 's magnitude does not order suppression either, and FoolsGold's is *positive*: **post-hoc and exploratory**, computed with no ASR and rescoring nothing (Appendices N, R). Two checks did pass: the $\kappa=0$ rung reproduces every standalone baseline exactly per seed, and `cos_krum`'s selection never moved (0/120).

1083 K.3 Mode A, the four aggregators, and the pre-registered second dataset

Mode A: the two regimes, and where they hold. **Mode A (payload-only)** closes the other channel: it keeps the benign weights uniform and sets the adversary-to-benign weight ratio to $r = e^\nu$ at fixed mean, so the adversarial share sweeps $0.049 \rightarrow 0.710$ with the statistic's dispersion held fixed: the mirror of Mode S. That gives Theorem 1 a single falsifiable shape: ASR single-peaked at $r=1$, since an over-attenuated adversary carries no payload (Regime B) and an over-amplified one is discarded by a rank aggregator (Regime A). It is **confirmed for coord_median**, the rank aggregator the regimes are stated for: ASR rises $0.124 \rightarrow 0.315 \rightarrow 0.443$ into $r=1$ and falls $0.443 \rightarrow 0.296 \rightarrow 0.098$ past it, both sides significant ($p_\downarrow = 3.0 \times 10^{-4}, 6.4 \times 10^{-4}$) at accuracy ≥ 0.74 (Figure 3). It is **refuted for reputation**, which is monotone, maximal at the most attenuated

1093 rung (0.894 at $r=0.135$), because an attenuated adversary reads as *consistent* to a reputation score.
 1094 Reputation is not a rank aggregator, so the shape was never licensed for it.

1095 **The same channels, across four aggregators of three structural kinds.** Across selectors on pair-
 1096 wise and on cosine distance (`krum`, `cos_krum`), a coordinate-wise order statistic (`coord_median`)
 1097 and a weighted averager with cross-round state (`reputation`), the negative is **not Krum-specific**: the
 1098 decision channel moves by 0.482–1.000 on three of the four while **the support of the admitted ad-**
 1099 **versarial mass changes in none of the 60 measured rounds of any of the four**, and on reputation
 1100 and `coord_median` that mass is *present* at baseline in 48 of those 60, so their counts are not floors
 1101 the way Krum’s is, and the only two arms whose ASR *rises* are the only two with nonzero $\Delta\Lambda_a$. **An**
 1102 **analysis of runs already reported, not a new experiment, and it orders nothing**: the larger $\Delta\Lambda_a$
 1103 (0.033) belongs to the arm with the *smaller* rise (Appendix C).

1104 **A second dataset and architecture, pre-registered.** The same cell on EMNIST-
 1105 byclass/simple_cnn in place of CIFAR-10/cifar_cnn, frozen in git before any EMNIST-byclass
 1106 ASR existed. Premise measured first, no ASR anywhere: the decision moves on 0.643–0.714 of
 1107 rounds, the support of the admitted mass unchanged in all 56 measured rounds (again a floor: no
 1108 adversary admitted at any rung). **Suppression shows no evidence of a practically meaningful**
 1109 **change**: $\Delta = -0.017$ against the frozen ± 0.15 margin, beside CIFAR-10’s -0.026 , at accuracy
 1110 $0.767 \rightarrow 0.835$, so the gate never binds. **Replicated on a second dataset and architecture**: the
 1111 frozen wording, never “generalizes”. Two limits from the freeze: the Jonckheere–Terpstra was
 1112 specified to catch a monotone *rise* and the trend is a *fall* ($z = -1.563$), so it is not read as support
 1113 for flatness; and with the identity rung at 0.027 the refuting branch was reachable (> 0.177) while
 1114 the indeterminate one was not, so equivalence is to the identity rung, not a low absolute ASR.

1115 It addresses neither recall nor discovery, and nothing in §5 is rescored.

1116 L Verification of the invariance mirrors

1117 All eight metric-swap labels were frozen in advance, and these mirrors are mechanism-
 1118 isolating ablations on the same footing as the instruments of §4. We check the statis-
 1119 tic directly rather than inferring invariance from an outcome (`verify_cos_invariance.py`,
 1120 `results/cos_invariance_check.json`), comparing each d_2 ’s internal state on $\{u_i\}$ against
 1121 $\{c_i u_i\}$ for the c_i NormClip and RFA actually produce. Over 18 live rounds (3 seeds $\times$ 2 up-
 1122 streams $\times$ 3 rounds) `cos_reputation`’s weight vector matches to $\leq 2.5 \times 10^{-7}$ with its ordering
 1123 never changing (0/18) and `cos_krum` selects the same client in 18/18; the Euclidean originals do
 1124 not (Reputation’s weights move by up to 0.222, ordering changes 12/18; Krum’s selection changes
 1125 6/18). Disturbance tracks the realized weight ratio ρ as Proposition 1 predicts: under NormClip
 1126 ($\rho \in [1.00, 4.90]$) Reputation changes in 3/9 and Krum in 0/9, under RFA ($\rho \in [2.11, 17.38]$)
 1127 in 9/9 and 6/9. On 1000 synthetic draws the separation is cleaner still (0/1000 vs. 992/1000
 1128 for the Reputations, 0/1000 vs. 618/1000 for the Krums), and the mirrors match the shipped im-
 1129 plementations they stand in for to $\approx 10^{-8}$ relative aggregate error. **The post-hoc within-defense**
 1130 **contrast of §5 re-uses this frozen suite**, so it is not pre-registered, and two seeds were added after
 1131 seeing the $n=3$ result: a top-up that cut both ways, degrading the invariant control’s Δ from ≈ 0
 1132 to $+0.19$ while leaving the disturbed/undisturbed split without overlap. One asymmetry matters:
 1133 `cos_reputation`’s *weights* are exactly invariant but its aggregate is not, since the weighted aver-
 1134 age is over the rescaled $c_i u_i$: a magnitude contraction of the Lemma 1 kind, so the prediction is
 1135 two-sided, with ASR staying low while clean accuracy pays a contraction cost.

1136 M Dose–response caveats recorded before the freeze

1137 **The pre-freeze disturbance measurement.** The abscissa was measured over 5 seeds $\times$ 3 rounds
 1138 per cell with no ASR involved, which is why it precedes the freeze: the class-(a) mirror’s cosine
 1139 scores move by at most 1.1×10^{-4} in 0/120 cells, i.e. the invariance holds in the shipped code
 1140 and not merely on paper, while the class-(c) arms reach 0.87 and 0.93. **Attack assignment.** The
 1141 class-(c) arms are on model-scaling and the class-(a)/(b) arms on the pixel backdoor: not a free
 1142 choice, since on model-scaling *no* invariant d_2 suppresses at usable accuracy (§5). “Same attack”
 1143 is required *within* each ladder, which is where the causal claim lives; across arms we compare

1144 curve shapes and slopes, never levels. `coord.median/pixel` sits only 0.057 below threshold, so
 1145 its binary above/below-0.5 reading is weak either way. The dose is uncorrelated with adversary
 1146 status by construction but not *adversary-adversarial*; a dose that deliberately up-weights adver-
 1147 saries would be strictly harder and is not run. A synthetic transform is not a real defense, so
 1148 we overlay the two *real* upstreams already measured on these cells (NormClip, RFA) as anchors
 1149 at their own measured disturbance; landing far off the synthetic curve would be a finding about
 1150 the instrument and reported as one. Design, grid, arms, shapes, margin and refutation criteria
 1151 are frozen in `pre_registration_dose_response.md` at commit e711a95, timestamped before
 1152 `results/dose_response/` existed and reproduced verbatim in the runner’s arm table. **What it**
 1153 **does not address:** recall, and discovery: `fg→rfa` remains a false negative of our own rule.

1154 N Dose–response ladder: per-seed detail, intervals, and the anchor residuals

Table 3: **The confounded ladder against the Mode-S instrument, on the one arm that has both.** Same downstream defense, same attack, same seeds, same nominal dose, and the two designs share their $\kappa=0$ cell *numerically* (mean and worst per-seed gap 0.00e+00, asserted at 10^{-12}). At the top rung they are matched on aggregate displacement, decision change and mean clean accuracy, so **the ladder’s fall is not an accuracy collapse**; they differ chiefly in the **adversarial coefficient channel**, and the ASR effect **changes sign**. **The one non-match, disclosed:** aggregate adversarial influence Λ_a moves in *both* designs, 0.081 against 0.033 ($2.4\times$), so neither column is a clean statistic-only intervention, which is why the contrast is “differ chiefly in the coefficient channel”, never “only in it”. *Vocabulary:* this row is the $\Delta\Lambda_a$ column of the channel table (the adversarial share of selected coordinates, the same field in the same artifact) and **not** its Δ adm. column, which counts rounds where the *support* of the adversarial mass changes and reads 0.000 on every row: what moves here is how much weight the adversary’s direction carries, not whether it enters at all. A two-column contrast on one arm: it locates where the two designs differ and attaches no *p*-value to the difference between the columns. Every cell recomputed from per-seed and per-round rows by `build_comparability_table.py`.

	Confounded ladder dose_kappa	Instrument (Mode S) doseS_kappa
downstream d_2	<code>coord.median</code>	<code>coord.median</code>
attack	committed pixel	committed pixel
seeds	42–46	42–46
nominal dose	κ 0.0 $\rightarrow$ 2.0, ρ 1 $\rightarrow$ 54.6	κ 0.0 $\rightarrow$ 2.0, ρ 1 $\rightarrow$ 54.6
benign coefficients	mean 1 over <i>all</i> clients	mean 1 over <i>benign</i> clients
adversarial coefficients	drawn from the same spread	pinned $c = 1.0$ exactly
adv. coefficient share	0.267 $\rightarrow$ 0.187	0.266667, constant
Δ_c	+0.000 $\rightarrow$ −0.336	0 by construction (read back 9e−08)
aggregate displacement	0.000 $\rightarrow$ 0.722	0.000 $\rightarrow$ 0.670
decision change	0.000 $\rightarrow$ 0.487	0.000 $\rightarrow$ 0.482
adversarial influence $\Delta\Lambda_a$	0.000 $\rightarrow$ 0.081	0.000 $\rightarrow$ 0.033
mean clean accuracy	0.767 $\rightarrow$ 0.690	0.767 $\rightarrow$ 0.698
ASR	0.443 $\rightarrow$ 0.170	0.443 $\rightarrow$ 0.541
ASR effect	−0.272 ($p_{\downarrow} = 0.0015$)	+0.098 ($p_{\uparrow} = 0.069$)

1155 Table 6 is every cell of the ladder; Figure 4 plots it on the measured axis and Figure 5 on the nominal
 1156 one. Both are produced by `analyze_dose_response.py` and `dose_response.figure.py` from
 1157 `results/dose_response/`; nothing below is transcribed from a log.

1158 **The replication arm’s post-hoc seed-matched difference.** The frozen primary for the
 1159 `coord.median` replication (§4) is the *unpaired* $\Delta = +0.098$, and it is not superseded by what
 1160 follows. Between-seed variance on this cell is large (identity ASR spread 0.287–0.677), which the
 1161 unpaired test cannot see through, so we also report the seed-matched difference: positive in 5/5
 1162 seeds, paired $t = 3.39$, one-sided $p = 0.014$. **This is post-hoc and was not pre-registered**; we
 1163 report it because it is the honest description of a small consistent effect, and we label it post-hoc
 1164 wherever it appears. It does not change the frozen verdict: the admission *ordering* is refuted either
 1165 way.

Table 4: Mode S (statistic-only), $n=5$ per arm ($n=8$ for `cos_krum`): mean ASR with the two measured channels. **The decision channel moves far more than ASR throughout.** [†] below the 0.35 accuracy gate. [‡] the replication arm, rank-discordant by design, frozen separately: $\Delta=+0.098$ is below reputation’s $+0.178$, so the cross-arm admission *ordering* is refuted while the negative replicates. Verdicts otherwise scored against readings frozen in advance; per-seed values in Appendix N.

arm	channel	$\rho=1$	2.72	7.39	54.6	test	verdict
krum / scaling	ASR	0.062	0.016	0.023	0.036	$p_{\uparrow}=0.70$	H-admission confirmed, H-stat. refuted
	decision	0.000	0.533	0.800	0.733		
	admission	0.000	0.000	0.000	0.000		
reputation / scaling	ASR	0.017	0.019	0.025	0.195	rise $+0.178$ vs. ladder $+0.762$ ($p=0.0017$)	H-admission confirmed (mostly atten.)
	decision	0.000	0.933	1.000	1.000		
	admission	0.000	0.002	0.008	0.020		
cos_krum / pixel	ASR	0.584	0.411	0.295	0.159 [†]	$p_{\downarrow}=8\times 10^{-4}$ $ \Delta =0.425$	refuted (Prop. 1 practical)
	decision	0.000	0.000	0.000	0.000		
	admission	0.000	0.000	0.000	0.000		
coord_median / pixel [‡]	ASR	0.443	0.442	0.490	0.541	rise $+0.098$ $< +0.150$ ($p_{\uparrow}=0.069$)	ordering refuted; negative repl.
	decision	0.000	0.232	0.358	0.482		
	admission	0.000	0.005	0.013	0.033		

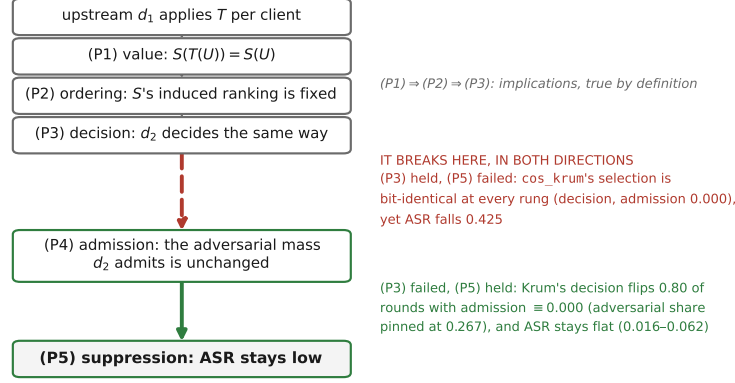
Table 5: The ladder as measured, $n=5$, per-attack mean ASR (mean clean accuracy). *Predicted* shapes were frozen before any run (Appendix X); $p_{\uparrow}$ is the one-sided Jonckheere–Terpstra probability of an increasing trend, the only direction any frozen rule tests. An arm meeting neither its confirmation nor its refutation criterion is reported *indeterminate* rather than scored in our favour. [†] below the 0.35 accuracy gate: a collapsed model, not suppression. Intervals, per-seed values, anchor residuals and both plotted axes: Appendix N.

arm (d_2 / attack)	predicted shape	mean ASR (clean acc.)				JT $p_{\uparrow}$	verdict
		$\kappa=0$	0.5	1.0	2.0		
1 krum / scaling	monotone rise	0.062 (0.57)	0.011 (0.65)	0.014 (0.68)	0.054 (0.62)	0.63	refuted
2 reputation / scaling	monotone rise	0.017 (0.78)	0.022 (0.78)	0.487 (0.77)	0.779 (0.70)	1.7×10^{-6}	confirmed
3 cos_krum / pixel	FLAT (control)	0.490 (0.61)	0.450 (0.57)	0.153 (0.47)	0.086 (0.22) [†]	0.999	indeterminate
4 coord_median / pixel	rise, shallower	0.443 (0.77)	0.428 (0.76)	0.327 (0.75)	0.170 (0.69)	0.998	refuted

1166 **The primary test, and two post-hoc sensitivities that cannot change its verdict.** The frozen
1167 primary test is a one-sided Spearman correlation of *rise* against *measured disturbance* pooled over
1168 all 16 cells, confirmed only at $\rho_s > 0$ with $p < 0.05$: we obtain $\rho_s = +0.351$, $p = 0.091$, so it
1169 is **refuted**. Two sensitivities are recorded because they are informative about *why*, and are labelled
1170 post-hoc because neither was frozen. (i) The four $\kappa=0$ cells are constructed (0, 0) points (zero
1171 disturbance and zero rise by definition) and dropping them gives $\rho_s = +0.780$, $p = 0.001$ ($n=12$);
1172 so the pooled correlation is carried by the arm-2 rise and destroyed by the two arms whose ASR
1173 falls, not by the identity anchors. (ii) Against the *nominal* dose κ (i.e. $\log \rho$) rather than measured
1174 disturbance, $\rho_s = -0.354$, $p = 0.911$: the wrong sign, which is the same fact as three arms falling.
1175 Neither is scored.

1176 **Per-arm trend tests.** One-sided Jonckheere–Terpstra across the four ordered rungs, 5 seeds each,
1177 $\alpha = 0.05$; the normal-approximation p agrees with a 20,000-draw permutation p to three decimals
1178 throughout. Arm 1: $J = 70$, $z = -0.34$, $p_{\uparrow} = 0.632$. Arm 2: $J = 144$, $z = +4.64$, $p_{\uparrow} =$
1179 1.7×10^{-6} . Arm 3: $J = 29$, $z = -3.10$, $p_{\uparrow} = 0.999$ ($p_{\downarrow} = 9.8\times 10^{-4}$); dropping the gated $\kappa=2.0$
1180 rung leaves $z = -2.06$, $p_{\downarrow} = 0.020$, so the decrease is not an artifact of the collapsed cell alone.
1181 Arm 4: $J = 31$, $z = -2.96$, $p_{\uparrow} = 0.998$ ($p_{\downarrow} = 1.5\times 10^{-3}$). No frozen rule tests the decreasing
1182 direction; $p_{\downarrow}$ is descriptive.

1183 **Clean accuracy against dose separates two ways of losing ASR.** Descriptive, not pre-registered.
1184 Accuracy trends by the same test: arm 1 $z = +1.35$, arm 2 $z = -3.30$, arm 3 $z = -3.50$, arm
1185 4 $z = -4.11$. Arm 3 pays $0.607 \rightarrow 0.220$: a 0.39 collapse through the Lemma 1 contraction
1186 channel, and the reason its $\kappa=2.0$ cell is uninterpretable rather than a success. Arm 4 pays only
1187 $0.767 \rightarrow 0.690$ while ASR more than halves, so there the dose genuinely *helps* coordinate median:
1188 a refutation of our predicted shape, but a benign mechanism. Arm 2 is the cleanest cell in the ladder,
1189 losing 0.076 accuracy while ASR rises 46 $\times$.



Statistic preservation is not security preservation: only the (P4) $\rightarrow$ (P5) link carries security content in our experiments.

Figure 2: **The preservation chain, and where it breaks.** (P1) $\Rightarrow$ (P2) $\Rightarrow$ (P3) holds by definition; the (P3)–(P4) link breaks in both directions, and both witnesses come from the oracle intervention that pins every adversary’s coefficient and so closes Lemma 1’s attenuation channel by construction (§4). The pooled statistic-level prediction we had frozen is separately **rejected** by a pre-registered ladder (§5), and a pre-registered replication in a different invariance class reproduces the negative but **refutes the cross-arm admission ordering**, so **admission is a direction, not a validated predictor**. Every number in the figure is recomputed from the scoring scripts’ own JSON by `story_chain_figure.py`.

Table 6: All 16 rungs, $n=5$. *disturb* is the independently measured disturbance of that arm’s own statistic at that rung, computed before the freeze with no ASR involved; *rise* is ASR relative to the arm’s own $\kappa=0$ rung and is the primary test’s ordinate. CIs are Student- t at $n=5$ and are wide by design: the ladder was powered to detect a monotone trend, not to resolve single cells. [†] fails the 0.35 clean-accuracy gate.

arm	κ	ρ	dist.	ASR	95% CI	rise	acc	per-seed ASR (42–46)					
krum	0	1.00	0.000	0.062	[0.019, 0.105]	+0.000	0.565	0.115	0.021	0.047	0.057	0.069	
scaling	0.5	2.72	0.600	0.011	[−0.001, 0.023]	−0.051	0.652	0.000	0.010	0.011	0.027	0.007	
class (c)	1.0	7.39	0.733	0.014	[0.003, 0.024]	−0.048	0.682	0.003	0.015	0.021	0.021	0.008	
	2.0	54.60	0.867	0.054	[0.002, 0.106]	−0.008	0.619	0.005	0.039	0.077	0.113	0.037	
reputation	0	1.00	0.000	0.017	[0.012, 0.022]	+0.000	0.777	0.014	0.012	0.017	0.023	0.019	
scaling	0.5	2.72	0.867	0.022	[0.014, 0.031]	+0.005	0.779	0.016	0.018	0.024	0.033	0.020	
class (c)	1.0	7.39	0.867	0.487	[0.349, 0.624]	+0.470	0.765	0.439	0.547	0.578	0.557	0.313	
	2.0	54.60	0.933	0.779	[0.673, 0.885]	+0.762	0.701	0.710	0.843	0.730	0.896	0.714	
cos_krum	0	1.00	0.000	0.490	[0.058, 0.922]	+0.000	0.607	0.003	0.269	0.630	0.864	0.686	
pixel	0.5	2.72	0.000	0.450	[0.130, 0.769]	−0.041	0.569	0.764	0.236	0.165	0.648	0.435	
class (a)	1.0	7.39	0.000	0.153	[−0.127, 0.433]	−0.337	0.466	0.132	0.090	0.000	0.543	0.001	
	2.0	54.60	0.000	0.086	[−0.100, 0.271]	−0.405	0.220 [†]	0.083	0.000	0.000	0.345	0.000	
coord_median	0	1.00	0.000	0.443	[0.263, 0.622]	+0.000	0.767	0.395	0.287	0.449	0.677	0.405	
pixel	0.5	2.72	0.251	0.428	[0.241, 0.615]	−0.015	0.764	0.454	0.215	0.430	0.638	0.402	
class (b)	1.0	7.39	0.378	0.327	[0.181, 0.472]	−0.116	0.749	0.351	0.139	0.352	0.463	0.328	
	2.0	54.60	0.487	0.170	[0.113, 0.228]	−0.272	0.690	0.179	0.090	0.204	0.192	0.187	

1190 **Anchor residuals, and why the instrument is the finding.** Residual is real-transform ASR
1191 ($n=5$, from `metric_swap_topup`) minus the synthetic rung at the nearest measured distur-
1192 bance. NormClip $\rightarrow$ Krum: disturbance 0/9, ASR 0.062 vs. rung 0.062, residual +0.001.
1193 RFA $\rightarrow$ Krum: 6/9 = 0.667, ASR 0.779 vs. rung 0.011 at disturbance 0.600, residual
1194 +0.767. NormClip $\rightarrow$ Reputation: 3/9 = 0.333, ASR 0.806 vs. rung 0.017, residual +0.789.
1195 RFA $\rightarrow$ Reputation: 9/9, 0.895 vs. 0.779, residual +0.116. Both `cos_krum` anchors sit at distur-
1196 bance 0 and land (+0.000, −0.044). `coord_median` was not in the metric-swap menu, so arm 4 has
1197 no real-transform anchor and its ecological validity is untested. The two large residuals are not a
1198 range mismatch: realized ρ is mean 3.61 / max 4.90 for NormClip and mean 11.93 / max 17.38 for
1199 RFA, both inside the ladder’s [1.00, 54.60] span. What differs is *which* clients get reweighted (Norm-

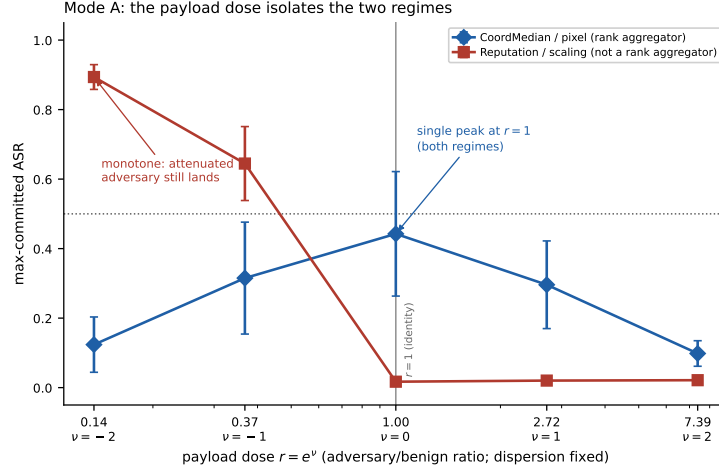


Figure 3: **The two regimes, exhibited: the payload dose separates them.** *The picture of regimes A and B of §2, produced by the instrument of §5.* With the statistic’s dispersion fixed and only the adversary/benign weight ratio r swept, `coord_median` (a rank aggregator, the class the regimes are stated for) is single-peaked at $r=1$ as Theorem 1 predicts, while reputation is monotone: off-spec, and the theorem is not claimed for it. Bars are Student- t 95% intervals, recomputed per seed by `analyze_targeted_dose.py`; the Mode-S companion is Table 4.

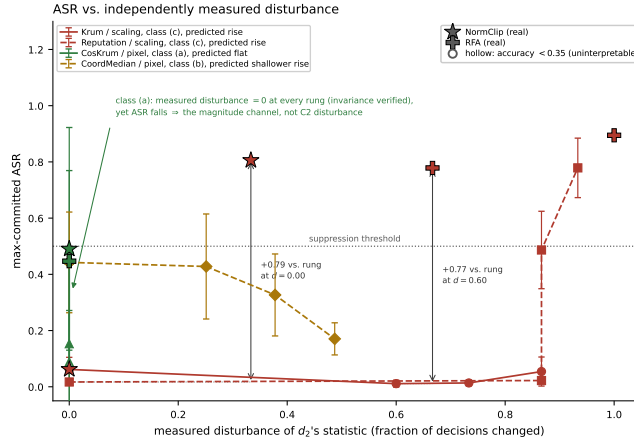


Figure 4: **The measured axis: ASR against disturbance of each arm’s own statistic**, an abscissa fixed before the freeze ($n=5$, Student- t 95% intervals; the figure script imports the arm and anchor construction from the scoring script, so the two cannot diverge). Stars and crosses are the two *real* upstream transforms measured on these same cells: at matched disturbance they sit $+0.77$ and $+0.79$ ASR *above* the synthetic curve, though their realized weight ratios (mean 3.61 and 11.93, maximum 4.90 and 17.38) lie *inside* the ladder’s span; §5 attributes the gap to Δ_c , post-hoc. The class-(a) arm’s four rungs coincide at disturbance exactly 0 and its ASR falls anyway: Lemma 1’s magnitude channel, not C2 disturbance. Hollow markers fail the 0.35 accuracy gate.

1200 Clip and RFA reweight in a way correlated with adversary status, our permutation deliberately does
1201 not) and we state that as the hypothesis caveat (iii) was written to test, not as a measured quantity:
1202 the shipped disturbance logs record only $c_{\min}$, $c_{\max}$ and ρ per round, not per-client coefficients, so
1203 an adversary-versus-benign coefficient contrast requires a new (cheap, no-ASR) measurement.

1204 **Two checks that passed.** The $\kappa=0$ rung reproduces each arm’s published standalone value exactly
1205 on every shared seed (16 seed-level comparisons across four arms, largest $|\Delta_{\text{ASR}}|$ and $|\Delta_{\text{acc}}|$ both
1206 $0.00\text{e}+00$ at tolerance 10^{-6}), so $\kappa=0$ really is the identity and the fresh rung and the published base-

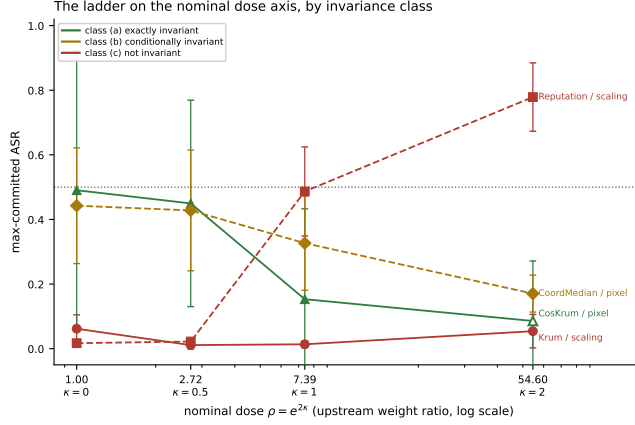


Figure 5: **The nominal axis:** the same 16 cells against the dose $\rho = e^{2\kappa}$, coloured by Proposition 1 class. Of the two class-(c) arms the fragility claim is stated over, only reputation rises; the other two arms *fall*. This is the axis the frozen per-arm shapes were stated on, and the pooled prediction was stated on the measured axis of Figure 4. Hollow markers fail the 0.35 accuracy gate.

line are the same quantity. **The first-rung fall we rest the (P4) $\nrightarrow$ (P5) witness on**, in full: `cos_krum / pixel` at $\rho=2.72$ falls 0.173 with clean accuracy moving only $0.599 \rightarrow 0.567$, paired $t=-1.976$, one-sided $p=0.044$, falling in 5/8 seeds. And `cos_krum`'s selection never changed in any of the 120 measured cells, with maximum relative score drift 1.1×10^{-4} : Proposition 1's classification of which statistics a positive per-client rescaling can disturb is verified directly, independently of the outcome the ladder then failed to predict.

O Secondary results: baselines, the adaptive stress test, scale, and switching

The margin behaves like a certificate, not just a switch. C3 is binary, but the margin m is measurable and its *strength* tracks security: over the 29 measured rounds with $w_a > 0$, higher margin goes with lower realized adversarial mass (Spearman $\rho = -0.48$, $p = 0.009$), Corollary 1's bound holding in every one, a number rather than a verdict, though the link is directional, not fitted, and post-hoc.

Table 7: Framework validation (representative pairs; full tallies in text). Threshold: LOW < 0.5, HIGH ≥ 0.5 .

Composition	Category	Predicted	Max ASR	Correct?
<i>PASS: C1 $\wedge$ C2 $\wedge$ C3 (predicted LOW)</i>				
FoolsGold $\rightarrow$ RFA	PASS	LOW	0.093	✓
FoolsGold $\rightarrow$ CoordMedian	PASS	LOW	0.107	✓
Reputation $\rightarrow$ CoordMedian	PASS	LOW	0.423	✓
<i>C1-FAIL: no constituent suppresses (predicted HIGH)</i>				
Reputation $\rightarrow$ RFA	C1-FAIL	HIGH	0.825	✓
NormClip $\rightarrow$ FoolsGold [†]	C1-FAIL	HIGH	0.887	✓
RFA $\rightarrow$ FoolsGold [†]	C1-FAIL	HIGH	0.914	✓
<i>C2-FAIL: discriminative property destroyed (predicted HIGH)</i>				
NormClip $\rightarrow$ Reputation	C2-FAIL	HIGH	0.819	✓
<i>C3-FAIL: margin compressed (predicted HIGH)</i>				
FoolsGold $\rightarrow$ TrimmedMean	C3-FAIL	HIGH	0.508	✓
NormClip $\rightarrow$ RFA	C3-FAIL	HIGH	0.853	✓
<i>False negatives (predicted HIGH, measured LOW)</i>				
NormClip $\rightarrow$ CM / RFA $\rightarrow$ CM	C3-FAIL	HIGH	0.433 / 0.485	×

[†]Reclassified from C2-FAIL: by Proposition 1(a) these preserve C2.

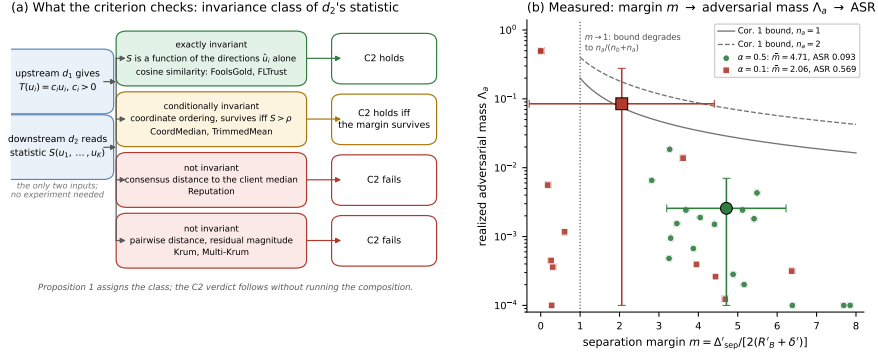


Figure 6: **The criterion is a mechanism test, and the mechanism is measurable.** (a) What C2 checks: the upstream transform T and the statistic S that d_2 reads determine an *invariance class* (Proposition 1), and the class determines the verdict with no composition run, which is why norm clipping does not blind FoolsGold but does blind Reputation. (b) The measured chain for FG→RFA, per round, on the transformed points RFA actually receives; each marker is one round with $w_a > 0$ (rounds where FoolsGold zeroes every adversary are Lemma 1 and carry no payload), large markers are mean $\pm$ s.d., and values below 10^{-4} are drawn at 10^{-4} . Corollary 1’s bound holds in every round, and the $\alpha=0.1$ collapse is visible in the theorem’s own quantity: m falls from 4.71 ± 1.51 ($n=17$) to 2.06 ± 2.35 ($n=12$), with several rounds at $m < 1$ where the bound degrades to $n_a/(n_b+n_a)$, and realized Λ_a rises $0.003 \rightarrow 0.085$ (max 0.50) as composition ASR goes $0.093 \rightarrow 0.569$.

1219 **Single-defense baselines.** Measured under the identical protocol (5 seeds, 50 rounds), no individual defense achieves low max-committed ASR: CM 0.52, FG 0.73, Rep 0.84, TM 0.85, RFA 0.89, NC 0.94. Against the best single defense (CM, 0.52), FG→RFA (0.093) is $5.6\times$ lower and FG→CM (0.107) is $4.9\times$ lower. FoolsGold alone reaches ASR 0.20 on model-scaling only by collapsing the model to 10% accuracy, so it is not suppression: consistent with its C1 failure.

1224 **Adaptive attacks against FG→RFA.** We evaluate one whitebox adaptive attack family that jointly evades both defenses: disjoint coordinate blocks minimize cosine similarity (evade FG) while ϵ -ball projection stays near the geometric median (evade RFA). The adversary has full state access, and these constraints are in tension. ASR against FG→RFA (5 seeds each, base quoted on the *same* five seeds so the comparison is seed-matched rather than against the $n=30$ mean): base committed 0.051 ± 0.011 ; Neurotoxin [Zhang et al., 2022] top-20% at scale 2.0 0.048 ± 0.009 ; projection only, no decorrelation, $\epsilon=2$: 0.050 ± 0.008 . With joint constraints the sweep over ϵ gives 0.036 ± 0.015 ($\epsilon=0.5$), **0.167 ± 0.069** ($\epsilon=1$), 0.130 ± 0.030 ($\epsilon=2$), 0.124 ± 0.025 ($\epsilon=4$). ASR peaks at $\epsilon=1$: tighter projection over-constrains the attack, larger ϵ fails to evade RFA, and without decorrelation FoolsGold suppresses it outright, so the competing constraints leave a narrow effective window. This is an empirical finding (Theorem 1 concerns mechanism preservation and does not bound an adaptive adversary whose optimization is untied to its conditions) and it probes one natural family of joint constraints at a handful of ϵ values, not the adaptive attack space. Within it, 0.167 is $3.3\times$ the seed-matched committed-attack ASR ($1.8\times$ the $n=30$ mean).

1238 **Scale.** At $N=100$, $f=0.05$, ResNet18 both pre-registered PASS predictions hold: FG→RFA max ASR = 0.069, FG→CM = 0.152. We flag that these runs reach only ≈ 0.29 clean accuracy at 50 rounds (ResNet18 is far from converged at this budget), so they establish that suppression transfers across scale and architecture, not that the composition is deployable at $N=100$ as configured.

1242 0.1 Heterogeneity, and where the criterion breaks

1243 Earlier drafts swept heterogeneity for single defenses only; no composition had been run at any $\alpha \neq 0.5$. We therefore ran all three PASS pairs across $\alpha \in \{0.1, 1.0, 10.0\}$ (3 seeds, both attacks, 50 rounds; max-committed ASR / clean accuracy):

Composition	$\alpha=0.1$	$\alpha=0.5$	$\alpha=1.0$	$\alpha=10.0$
FG→RFA	0.569 / 0.20	0.093 / 0.50	0.051 / 0.57	0.063 / 0.62
FG→CM	0.433 / 0.40	0.107 / 0.68	0.133 / 0.74	0.129 / 0.78
Rep→CM	0.548 / 0.48	0.423 / 0.76	0.377 / 0.79	0.326 / 0.81

In our sweep, which is three seeds per cell and fixes direction rather than magnitude, the certified compositions stay below threshold for $\alpha \in \{0.5, 1.0, 10.0\}$, including the near-IID setting where one might expect FoolsGold to founder on legitimately similar benign updates: $\alpha=10$ is indistinguishable from $\alpha=0.5$. **Under extreme heterogeneity ($\alpha=0.1$) two of the three PASS compositions cross the threshold** (FG→RFA 0.569, Rep→CM 0.548) and only FG→CM survives, at 0.433: a real boundary condition on the framework, with caveats in both directions: FG→RFA at $\alpha=0.1$ is strongly bimodal across seeds (0.065–0.923), so at $n=3$ the direction is clear but the magnitude is not, and clean accuracy degrades badly (0.20–0.48), so the regime stresses utility as well as security. Crucially Theorem 1 supplies a diagnostic quantity that retrospectively explains this boundary: re-measuring the transformed points at $\alpha=0.1$, RFA’s separation margin $m := \Delta'_{\text{sep}}/[2(R'_B + \delta')]$ collapses from an all-round mean of 3.24 (met in 68% of rounds) to 1.01 (met in 20%), over the payload-carrying rounds alone, the ones Figure 6 plots, the same collapse reads 4.71 ± 1.51 ($n=17$) to 2.06 ± 2.35 ($n=12$) with the per-round minimum falling $2.82 \rightarrow 0.00$, the difference being that the all-round means average in the Lemma 1 rounds at $m=0$, exactly the boundary $m \rightarrow 1$ where Corollary 1’s bound degrades to 40% adversarial mass, because extreme heterogeneity inflates the benign radius R'_B until the benign cluster is no longer separated from the adversarial points, which is the precondition the theorem requires. A practitioner running the 5–10 round pilot we recommend would measure $m \approx 1$ and decline to deploy. The criterion is therefore mechanistically diagnostic at this boundary (the property that makes a screening rule usable) though this is a post-hoc explanation of a known failure, not a pre-registered prediction of it.

O.2 Composition vs. randomized switching

High persistence sharply reduces the value of temporal switching. In our tested regime, $\gamma \approx 0.66$ ($\gamma := \text{ASR}(t+1)/\text{ASR}(t)$ after one clean round): triggers decay slowly enough that intermittent suppression cannot clear them, so temporal mixing approaches the weaker constituent defense’s performance. Whether switching retains any value depends jointly on γ , injection strength, switching frequency, and recovery dynamics: persistence alone does not imply switching is worthless. The quantitative boundary is architecture-specific; the qualitative prediction (high- γ attacks blunt switching) is what this secondary analysis contributes, and nothing in the preservation argument rests on it.

Order-sensitivity control. Both variants apply both defenses every round and differ *only* in order, so compute is matched and any gap is structural. We report it as a **motivating demonstration that composition order is structurally consequential, not as evidence for the criterion**: the mechanism behind the largest gap is a degeneracy rather than a signal-preservation failure, as the diagnosis below makes clear. Pixel ASR, 5 seeds: fixed FG→CM 0.076 ± 0.035 versus randomized 50/50 ordering 0.681 ± 0.133 ($9\times$) and temporal mixing of one defense per round 0.679 ± 0.159 ; fixed FG→RFA 0.037 ± 0.011 versus randomized 0.719 ± 0.113 ($19\times$). The mechanism is the degeneracy of §1: CoordMedian supplies no per-client transform, so every CM→FG round reduces to FoolsGold alone, leaving half the rounds defended by a single constituent that cannot suppress on its own. On a reputation+TrimmedMean menu ($n=30$), composition reaches ASR 0.316 against 0.915 for temporal mixing.

P The metric-swap suite’s two-sided prediction

Because C1 binds for all eight pairs, every frozen label in the metric-swap suite was HIGH (a consequence of the measured inputs rather than a later choice), so the suite bears on specificity only. One prediction inside it was genuinely two-sided. `cos_krum`’s C1 verdict on model-scaling turns on whether the accuracy gate is applied (0.078 ASR at 0.146 clean accuracy), so we froze *both* readings in advance: primary (gated) $\Rightarrow$ HIGH, secondary (numeric only) $\Rightarrow$ LOW, with the rule fixed beforehand that a LOW outcome at accuracy ≥ 0.35 would mean the gate was wrong. Both cells came out **HIGH** (0.508, 0.566), so the gate is vindicated against a pre-registered alternative

that would have overturned it. This is the only place in the paper where our own accuracy gate was put at risk by a frozen rule.

Q The prospective suite, arm by arm

(i) **FLTrust as d_1 (7 pairs, 2/7): out of scope.** When d_1 alone suppresses, the composition inherits that suppression whatever d_2 does, so all five predicted-HIGH pairs are misses even though their C2/C3 failures are real: FLTrust genuinely destroys Reputation’s consensus-distance signal, but destroying a mechanism that was never load-bearing costs nothing. This is C0, and it was invisible on the original 42 pairs, where the strongest single defense reaches only 0.519 and the lowest ASR among all 39 non-PASS pairs is 0.433; no pair was ever out of scope. We apply C0 here *retrospectively*, which is weaker than a pre-registered exclusion.

(ii) **FLTrust as d_2 (6 pairs, 6/6): in scope, but nearly an identity.** Proposition 1(a) predicts $\cdot \rightarrow \text{FLTrust}$ preserves C2, and it does, but FLTrust gates on a scale-invariant cosine and then *renormalizes every update to $\|g_{\text{STV}}\|$* , discarding magnitude outright, so a positive rescaling $u_i \mapsto c_i u_i$ is *exactly* cancelled: FLTrust discards precisely what d_1 modifies. The per-seed data show this directly. $\text{Krum} \rightarrow \text{FLTrust}$ and $\text{MKrum} \rightarrow \text{FLTrust}$ are **bit-identical** to FLTrust alone on all six runs (pass-through d_1); $\text{NC} \rightarrow \text{FLTrust}$ is bit-identical *on the pixel attack*, where no client exceeds $\tau=5$ so $c_i \equiv 1$, and differs only within seed noise on model-scaling, where clipping fires; $\text{Rep} \rightarrow$ and $\text{RFA} \rightarrow \text{FLTrust}$ likewise differ only within noise. The lone real deviation is $\text{FG} \rightarrow \text{FLTrust}$ (0.064, up to 0.084 per seed): the one d_1 producing exact *zeros* rather than positive rescalings, changing FLTrust’s client set instead of its magnitudes, which is the Lemma 1 interaction. So this arm confirms that theory and implementation agree, and demonstrates the degeneracy mechanism cleanly, but it is **not falsifiable**: every d_1 in our menu is a *non-negative* per-client rescaling, FLTrust cancels the positive part exactly, and so the only channel left open is the *zero pattern*, which is precisely where the one deviation appears. Proposition 1(a) permits no other outcome on the strictly-positive part, and Lemma 1 accounts for the rest.

(iii) **Neither defense is FLTrust (7 pairs, 5/7): the only arm that could have surprised us.** All were predicted HIGH; correct are $\text{rep} \rightarrow \text{krum}$ 0.880, $\text{nc} \rightarrow \text{mkrum}$ 0.943, $\text{rfa} \rightarrow \text{mkrum}$ 0.875, $\text{krum} \rightarrow \text{rfa}$ and $\text{mkrum} \rightarrow \text{rfa}$ both 0.870. Neither miss is a mechanism error: $\text{fg} \rightarrow \text{krum}$ reads 0.078 at 9.9% clean accuracy (a collapsed model, the same hollow-suppression artifact as FoolsGold alone) and $\text{krum} \rightarrow \text{cm}$ reads 0.450 against a 0.5 threshold as a degenerate pair reducing to CoordMedian alone, whose own baseline is 0.519; threshold noise on a borderline constituent. This arm contains *zero* predicted-LOW pairs, so like Tier 2 it bears on specificity only.

The suggestive contrasts, and why they are still confounded. Three matched pairs hold d_1 fixed while varying only d_2 ’s invariance class: $\text{rep} \rightarrow \text{fltrust}$ 0.046 vs. $\text{rep} \rightarrow \text{krum}$ 0.880; $\text{nc} \rightarrow \text{fltrust}$ 0.046 vs. $\text{nc} \rightarrow \text{mkrum}$ 0.943; $\text{rfa} \rightarrow \text{fltrust}$ 0.047 vs. $\text{rfa} \rightarrow \text{mkrum}$ 0.875: a 19–20 $\times$ gap turning on exactly the property Proposition 1 identifies. They are nonetheless confounded in the way Lemma 2 predicts: the invariant-signal d_2 is also the only d_2 in this menu that suppresses alone, so effectiveness and mechanism co-vary across the contrast. This is why breaking the confound requires holding standalone effectiveness fixed rather than choosing a different defense.

R Adversary-correlated dispersion Δ_c , measured

R.1 Heterogeneity is not the mechanism: ρ against Δ_c

The finding, stated first. What costs a downstream defense its suppression is not how heterogeneous the upstream transform is but **whether its reweighting is correlated with adversary identity**, and not even the size of that correlation, which has **two opposite regimes**. This is why our own headline predictor failed.

Two different things a coefficient vector does, and the theory sees only one. (Symbols: Appendix G.) Proposition 1 turns on **dispersion**, $\rho := \max_i c_i / \min_i c_i$ (how far apart the coefficients are, which decides whether an ordering survives) and says nothing about *which* clients receive which coefficient. That second property is **adversary-correlated dispersion**,

$$\Delta_c := \mathbb{E}[c_i \mid i \in A] - \mathbb{E}[c_i \mid i \notin A], \quad A \text{ the adversarial participants,}$$

and the two are **independently controllable**, not merely uncorrelated in our sample: a transform can spread the coefficients arbitrarily at $\Delta_c = 0$ exactly, or leave them nearly equal while shifting the adversary’s share, and Proposition 3 exhibits a family of each. **Proposition 1 and Theorem 1 constrain ρ ; Lemma 1’s attenuation channel is about Δ_c** , and §5 finds the real transforms differ on the second. Δ_c is *not* a screening quantity: computing it requires knowing A (§6).

Two mechanism-preserving regimes, at opposite ends of ρ . Lemma 1 is a second, independent route to suppression, and the two together explain all three PASS compositions. Let $r := w_a / \max_b w_b$ and S be the attack’s coordinate extremeness. **(A) Ordering preservation** keeps the adversary in the tail where a rank aggregator discards it ($S \cdot r > 1$). **(B) Attenuation** instead drives the adversarial transformed value *inside* the benign spread, where it cannot be an extreme order statistic and its payload is scaled by r ($S \cdot r \ll 1$; $r = 0$ is exact annihilation). The hazard is $S \cdot r \approx 1$, and our two weighting defenses sit at opposite ends: FoolsGold’s weights are bounded ($\rho \leq 2.1$, $S \cdot r \geq 4.8$), regime A; Reputation’s are wildly unbounded (ρ mean 1.4×10^5) precisely *because* it crushes the adversary ($S \cdot r = 0.017$), regime B. So Reputation→CoordMedian violates the ordering condition by five orders of magnitude and still suppresses (Figure 3, App. N, exhibits both regimes). Hence ρ does not order Δ_c (Proposition 3), and Δ_c ’s *magnitude* does not order suppression either, the same large $|\Delta_c|$ being protective at one end and catastrophic at the other. **Neither quantity is a predictor**; they buy an account of *why* a heterogeneity-based reading mispredicts.

$\Delta_c := \mathbb{E}[c_i \mid i \in A] - \mathbb{E}[c_i \mid i \notin A]$ for every per-client transform in the paper, over the same 5 seeds $\times$ 3 rounds and, within a round, the *same* raw update stack for every transform, so the rows are comparable by construction (measure_coefficient_targeting.py, results/coefficient_targeting.json). **No ASR is computed and no model is trained per transform**, which is what lets this be added without rescoring anything; the figures are **post-hoc and exploratory** and no frozen prediction depends on them. Coefficients are read back from transformed norms, $c_i = \|T(u_i)\|/\|u_i\|$, so Table 8 measures the shipped code path rather than the builder’s description of it. As a provenance check the synthetic families’ adversarial coefficient shares reproduce results/admission_measurement.json to 0.0 absolute on all 13 cells, so these rows may be read on the same axis as the Mode-S and Mode-A figures of §4.

Table 8: ρ says how much a transform spreads the coefficients; Δ_c says who receives the spread. They are not ordered together: RFA and NormClip target the adversary 2–3.6 $\times$ harder than the ladder’s most dispersed rung at ρ that is 3.7–13 $\times$ *smaller*. Means over non-degenerate rounds; the synthetic rows are attack-pooled since their coefficients do not depend on the attack.

transform	attack	ρ	$\mathbb{E}[c \mid A]$	$\mathbb{E}[c \mid \neg A]$	Δ_c
NormClip	model-scaling	4.19	0.268	0.989	−0.721
NormClip	pixel	1.00	1.000	1.000	0.000
Reputation	model-scaling	8.3×10^5	0.003	1.387	−1.383
Reputation	pixel	1.68	0.915	1.019	−0.105
FoolsGold	model-scaling	1.9×10^{12}	1.497	0.834	+0.663
FoolsGold	pixel	1.8×10^{12}	1.365	0.986	+0.378
RFA	model-scaling	14.8	0.125	1.332	−1.208
RFA	pixel	1.69	0.923	1.012	−0.089
<i>Synthetic families of §5–§4, same code path:</i>					
ladder, $\kappa=0.5/1.0/2.0$	—	2.72/7.39/54.6	—	—	−0.140/−0.240/−0.336
Mode S, all rungs	—	1.00–54.6	—	—	0.000
Mode A, $\nu=-2 \dots +2$	—	7.39–7.39	—	—	−1.138 $\dots$ + 2.469

Three readings, in decreasing confidence. **(1) ρ is not a proxy for Δ_c** : the two are anti-ordered on the anchors, and NormClip on the pixel backdoor has $\rho = 1$ exactly because no client exceeds $\tau=5$, so $\Delta_c = 0$ there for a reason the invariance classification cannot see. **(2) The two anchors that miss the ladder are the two most extremely targeted**, which explains the residuals of §5 without appeal to a range mismatch. **(3) Δ_c is nonetheless not a predictor of suppression**: RFA is the most negative of the four real transforms and rfa→krum fails at 0.779, while Reputation is more negative still and rep→coord.median suppresses. That is the two-regime structure of §2 again: strong negative Δ_c can mean either annihilation or mere dilution, and Δ_c ’s magnitude alone does

not say which. We report it as the axis on which the ladder’s anchors were unmatched, not as the quantity that replaces the statistic.

S Confusion matrix in full

Exactly 5 of the 42 pairs have max-committed ASR < 0.5 : the 3 PASS pairs plus the 2 false negatives. Pooling both in-distribution tiers (each pair scored against the label committed for it at the time) gives TP = 3, FP = 2, TN = 35, FN = 2 (38/42; specificity 94.6%, which we report but do not lead with, since 37 of 42 pairs are negatives and specificity is inflated by that imbalance; precision 1.0 at base rate 11.9% is the $8.4\times$ enrichment that matters, and recall is 40% under the tightened C1, since FG→RFA now counts as a miss). We report this pooled figure with its decomposition rather than on its own, because the tiers are not equivalent evidence: it is 16/18 development *fit* plus 22/24 held-out *prediction*. The four sub-tiers the body refuses to pool are development *fit*, held-out *specificity*, out-of-sample *positives with no informative negatives*, and the 13/20 *prospective* suite. Scoring instead against the criterion’s mechanism labels, which classify both failed LOW calls as HIGH, gives TP = 3, FP = 0, TN = 37, FN = 2 (40/42); we take the more conservative pooled figure as primary. The two false positives, fg→NC (0.927) and fedavg→CM (0.519), were hedged as speculative at pre-registration time and neither satisfies C1; every pair that actually satisfies $C1 \wedge C2 \wedge C3$ came out LOW. Category tally after the Proposition 1 correction: degenerate 18, C1-fail 11, C2-fail 3, C3-fail 7, PASS 3, and all 3 surviving C2 failures are $\cdot \rightarrow$ Reputation, exactly the non-invariant class of Proposition 1(c). With a positive set of only 2 certified in-distribution pairs and recall 40%, the 94.6% specificity figure is high partly because the positive class is rare, so we report it as a supporting statistic rather than a headline. The conditions are sufficient, not necessary, and they do reject working compositions. Extension to 3-way compositions: 9/10 triples correct (too small for confidence).

T Diagnosing the two false negatives in full

NormClip→CM (0.433) and RFA→CM (0.485) were labelled C3-FAIL on the reading that *any* heterogeneous upstream rescaling compresses a rank aggregator’s margin. Theorem 1(1) says something weaker and quantitative: ordering survives when the attack’s coordinate extremeness exceeds the realized weight ratio, $S > \rho$. Measured, NormClip’s ratio is mild. Over 9 live rounds it clips only 0–2 of 5 clients (11 of 45 client-rounds), so $\approx 76\%$ of clients receive $c_i = 1$ *exactly*, and the largest raw norm we observe is 24.5 against $\tau=5$, bounding $\rho \leq 4.9$; against $S=10$ the margin $S/\rho \geq 2.0$ still clears 1. RFA-reweighting is likewise near-uniform on benign clients. Both pairs therefore sit close to the *uniform* rescaling case, where ordering is preserved exactly, which is why their ASR lands in an intermediate band just under threshold rather than in the high band the binary label predicts. **The refinement does not recover them.** Both also fail C1: CoordMedian alone is already at 0.519, and neither NormClip (0.935) nor RFA (0.885) suppresses, so no reading of C3 makes either pair certifiable. What the diagnosis buys is a sharper condition and an account of the residual: a d_2 whose own baseline sits at 0.519 needs only a mild order-preserving upstream transform to cross 0.5, so these two are threshold noise around a marginal aggregator, not suppression the framework failed to recognise. The operational refinement is to evaluate C3 for rank aggregators against the *realized* ρ from a 5–10 round pilot rather than the binary question of whether d_1 is heterogeneous: the same quantity Theorem 1(1) already requires, which the labelling had not been using.

U Condition-by-condition scoring, and the run-count saving

The results below are §3’s, moved here for space. None changes a reported number.

The out-of-sample bookkeeping, in full. All 4 out-of-sample PASS predictions the tightened criterion licenses held, as did both in-distribution $C1 \wedge C2 \wedge C3$ pairs. **But all 7 are positive predictions whose FAIL controls are uninformative** (DBA fails even undefended), which is why §3 declines to read them as an out-of-sample classification result: five positive predictions against an 11.9% base rate (5/42, Appendix S), with the positive direction separately unvalidated (§4).

1430 **Does every condition earn its place? C3 does not, reported as a finding.** Scoring all 42 pairs
 1431 on each condition *independently*: C1 alone selects 4 at 50% precision, C2 alone 14 at 36%, C1∧C2
 1432 selects 2 at 100%, and **C3 alone selects 35 at 9%**, below the 12% base rate, never binding on a
 1433 pair C1∧C2 already admits. So **we demote C3 to a margin diagnostic and recommend C1∧C2**,
 1434 retaining it on theoretical grounds only: Theorem 1 makes margin compression a real failure mode
 1435 and the $\alpha=0.1$ failure of §3 is exactly a margin collapse. It changes no number: a finding about the
 1436 criterion, not an improvement to it.

1437 **What the saving is.** Counted in runs rather than pairs (C1’s standalone evaluations are runs too)
 1438 the saving is 420 **unscreened against 80 screened**, 81% **fewer composition-evaluation runs**, not
 1439 the 95.2% of *pairs* we previously reported; **nothing is eliminated before any run**, and the structural
 1440 claim is the scaling $|A| S D(D-1)$ (*quadratic* in the defense count) to $|A| S ((D-1) + |\text{admitted}|)$.
 1441 GPU-hour figures are *estimates*; no timing telemetry was recorded.

1442 V What the boundary permits, claim by claim

1443 The inventory below separates the three kinds of *no* this paper reports: *structural* (no design satis-
 1444 fying the gate can answer the question, Prop. 4), *open* (answerable in principle, but not by an oracle
 1445 instrument), and *scope* (outside what the committed-attack setting measures). Nothing here is a new
 1446 claim; each row points at the section that establishes it.

Table 9: **What this design can and cannot conclude.** The middle column is about *identifiability*, not statistical power: a “no” marked *structural* cannot be repaired by more seeds, defenses or attacks (Proposition 4). Every entry is scoped to the measured setting of §5.

question	identifiable?	answer, and its evidence
Does disturbing d_2 ’s statistic change its <i>decision</i> ?	measured	Yes: Krum’s selection flips in up to 0.80 of rounds
Does it change the <i>admitted adversarial mass</i> ?	by construction	No: unchanged in every one of the 60 measured rounds
Does it change <i>suppression</i> ?	instrumented	No: $ \Delta = 0.026$, inside margin (Prop. 2(a))
Does <i>adversary-correlated</i> reweighting change suppression?	instrumented	Yes , non-monotonically: single-peaked (Mode A)
Does admitted mass <i>order</i> outcomes across defenses?	pre-registered	No: refuted: $+0.098$ against a frozen $+0.150$
Is (P4) <i>sufficient</i> for (P5)?	control arm	No: <code>cos_krum</code> admits 0.000 while ASR falls
Can (P4) be screened <i>without</i> adversary identity?	no: open	Both instruments are oracles: the stated open problem
Does C1 identify C2’s contribution?	no: structural	Empty cell by the gate, for every menu (Prop. 4)
Does any of this certify <i>deployed</i> security?	no: scope	Committed attack only; one adaptive test breaks it

1447 W Anticipated objections

1448 Five objections we expect, answered against text that is already in the paper rather than by new
 1449 argument. Nothing here is a new claim; each answer is a pointer plus the scope it turns on.

1450 **1. This is small-scale federated learning, not decentralized foundation-model training.**
 1451 Agreed, and we claim **methodological** relevance only (§1): no transformer arm, no foundation-
 1452 model measurement, and no sentence implying either. What transfers is a statement about *causal*
 1453 *structure*: any criterion whose eligibility gate is defined on the same outcome it predicts conditions
 1454 on a descendant of both mechanism and effectiveness (Prop. 4), and that argument does not depend
 1455 on model scale or participant count. What does *not* transfer is every empirical magnitude, which
 1456 is why each is reported per arm, tiered at **very unequal weight and never pooled** (Table 2), and
 1457 scoped in Table 1. The result forbids an inference; it does not measure a system.

1458 **2. The interventions are oracles: no deployed system can pin an adversary’s coefficient or**
 1459 **read adversary identity.** Correct, and that is why Mode S is an **instrument, not a screen**. The
 1460 Conclusion says so in those terms: Mode S measures level (P4) only by reading adversary identity,
 1461 which no deployable defense can, so our experiments **identify the causal channel and are not a**
 1462 **(P4) screen**. Pinning $c=1.0$ has one purpose, to cut the confounded path of Figure 1(a) by construc-
 1463 tion, exactly as an oracle-labelled control isolates a channel in an ablation. The deployable question
 1464 is left open as the paper’s closing sentence. The one control that *removes* an oracle component

1465 (score-only, which scores the transformed stack but aggregates the untransformed selected update)
 1466 does not change the verdict (App. E, D).

1467 **3. $n=5$ seeds is small.** It is: $n=5$ per arm, $n=8$ for `cos_krum`, with Student- t 95% intervals
 1468 throughout and no claim that a small effect is zero. Where suppression does not move we report **no**
 1469 **evidence of a practically meaningful change under the pre-specified ± 0.15 equivalence margin**,
 1470 a margin frozen before the runs (Table 1), never a claim that suppression is unchanged. Two design
 1471 choices do more for precision than seeds would: the identity rungs are imported **bit-identical per**
 1472 **seed** (largest deviation 0.00e+00), so the contrast is within-seed rather than across noise, and the
 1473 central claim is a **sign reversal** (-0.272 against $+0.098$ on the identical cell, Table 3), which is not
 1474 a null result. Finally, Prop. 4 says the cell that would validate the criterion is *empty by design*: more
 1475 seeds cannot repair it (Table 9).

1476 **4. If ASR does not move, why does a change of decision matter?** Because the change of de-
 1477 cision is what a preservation criterion *measures*, and the gap between the two is the finding. An
 1478 invariance argument certifies the top three levels of Def. 1 while the security content sits at level
 1479 (P4), the adversarial contribution admitted. A criterion reading the Krum arm would call that com-
 1480 position broken (the decision flips in up to 80% of rounds) for a composition whose suppression
 1481 shows no measurable change: a false alarm in the direction that costs coverage (Prop. 5, recall
 1482 bound $1/5$). The converse fails too: `cos_krum`’s selection is bit-identical at every rung yet it never
 1483 suppresses (0.584 at the identity) and its ASR *falls* 0.425 across the ladder, so exact invariance of
 1484 the statistic neither implies suppression nor holds it fixed (App. J.5).

1485 **5. The theory covers only positive per-client rescalings.** Yes, and we say so wherever it is used:
 1486 **a laboratory for studying composition, not a general model of it** (§1; Prop. 1, Thm. 1, App. I).
 1487 The class is not arbitrary (it is what norm clipping, contribution weighting and reputation weighting
 1488 do to a client’s update) but it excludes coordinate-wise filters and any transform that changes an
 1489 update’s direction, and the theorem’s own margin closes under heterogeneity ($m=3.24 \rightarrow 1.01$ at
 1490 $\alpha=0.1$), so **it certifies nothing where heterogeneity is high**. The negative does not rest on the
 1491 theory: it is a *measured* counterexample, and a narrower theory makes the counterexample sharper
 1492 rather than weaker, since **the theorem certifies mechanism preservation, not security**, it bounds
 1493 no ASR.

1494 X Experimental Details

1495 Figure 7 places the evaluated settings in the (γ, p_{eff}) plane. At the persistence we measure ($\hat{\gamma} \approx$
 1496 0.66) the one temporal mix we ran reaches ASR 0.956, which is the reason the paper composes
 1497 rather than switches; the figure records those points and draws no boundary between them.

1498 **FL configuration.** $N=10$ clients, $K=5$ per round, 50 rounds, SGD with momentum 0.9, learning
 1499 rate 0.01, local epochs 3, batch size 32, Dirichlet $\alpha=0.5$ (non-IID). Scale experiments: $N=100$,
 1500 $K=10$, $f=0.05$, ResNet18 with GroupNorm.

1501 **Attacks.** Model-scaling: scale factor 10, 50% poison fraction, target class 0. Pixel backdoor:
 1502 5×5 bottom-right trigger, target class 0. DBA: 4 sub-triggers at image corners, $f=0.4/0.6$. Joint-
 1503 constraint: $\epsilon \in \{0.5, 1.0, 2.0, 4.0\}$, disjoint coordinate blocks per adversary, single-step ϵ -ball pro-
 1504 jection per round.

1505 **Defenses.** NormClip $\tau=5.0$, i.e. $u_i \mapsto u_i \min(1, \tau/\|u_i\|_2)$ over the whole flattened update;
 1506 TrimmedMean $\beta=0.2$ (trim 20% each tail); RFA: unweighted Weiszfeld, 50 iterations, tolerance
 1507 10^{-6} , initialised at the mean; FoolsGold: $w_i \propto \log((1 - s_i)/(s_i + \epsilon) + \epsilon)$ with s_i the client’s
 1508 maximum absolute pairwise cosine similarity normalised by the round maximum, $\epsilon=10^{-5}$, then
 1509 shifted by $-\min_i$ and normalised, so the least-trusted client of each round receives weight exactly
 1510 0 (empirically 2 of 5 clients per round).

1511 **Reputation.** Despite the name this defense is *memoryless*, it keeps no cumulative per-client state
 1512 and is recomputed from scratch each round from that round’s K participants. Consensus is the
 1513 coordinate-wise median m of the K updates; scores are ℓ_2 distances $d_i = \|u_i - m\|_2$; the bandwidth

1514 is adaptive, $\sigma = \text{median}_i(d_i)$ (clamped $\geq 10^{-6}$); weights are $w_i = \exp(-d_i/\sigma)$, normalised to sum
 1515 to 1. No clipping, no initialisation. Its discriminative property is therefore *consensus distance*, not
 1516 similarity spread, which is why Proposition 1(c) predicts that $\cdot \rightarrow \text{Reputation}$ compositions fail C2.
 1517 Weights are strictly positive, so Reputation never produces the zero vectors of Lemma 1.

1518 **Adversary count per round.** Adversaries are the fixed client identities $\{0, \dots, fN - 1\}$; each
 1519 round samples K participants uniformly without replacement from all N , with no stratifica-
 1520 tion. For $N=10$, $f=0.2$, $K=5$ the per-round adversary count is hypergeometric: $P(n_a=0) =$
 1521 $P(n_a=2) = 56/252 = 0.222$, $P(n_a=1) = 140/252 = 0.556$; measured over 100 rounds we ob-
 1522 serve $0.19/0.54/0.27$. Two consequences, both matching the data. CoordMedian’s count majority
 1523 $n_a < K/2 = 2.5$ holds for every attainable $n_a \in \{0, 1, 2\}$, i.e. in 100% of rounds. TrimmedMean’s
 1524 budget is $\lfloor \beta K \rfloor = \lfloor 1.0 \rfloor = 1$, so it is exceeded whenever $n_a=2$: 22.2% of rounds, ≈ 11 of 50.
 1525 This is why FG $\rightarrow$ TrimmedMean fails (0.508) while FG $\rightarrow$ CoordMedian succeeds (0.107) under an
 1526 otherwise identical upstream transform. We retain the pre-registered C3-FAIL label for FG $\rightarrow$ TM;
 1527 the count-budget violation is an additional mechanism, not a relabelling.

1528 **Theorem quantities.** FoolsGold weight ratio across 5 seeds $\times$ 20 rounds ($N=10$, $f=0.2$, model-
 1529 scaling): $\max \rho = 2.1$, giving extremeness ratio $S/\rho \geq 4.8$ with $S=10$ (regime A). Reputation,
 1530 same protocol (5 seeds $\times$ 20 rounds, 81 rounds containing both adversarial and benign clients): ρ
 1531 mean 1.44×10^5 , max 2.84×10^6 , p95 6.1×10^5 ; mean adversarial weight 4.83×10^{-4} against mean
 1532 benign weight 0.278, and $w_a < 10^{-3}$ in 85% of rounds. Reputation therefore fails the ordering
 1533 condition by ~ 5 orders of magnitude and suppresses via regime B instead. RFA, measured on
 1534 the transformed points $u'_i = w_i u_i$ that RFA actually receives (25 adversary-present rounds over 3
 1535 seeds): FG zeroes every adversary in 32% of rounds, where Lemma 1 applies and the measured
 1536 contraction $1 - \Lambda_a$ is 0.29 on average (range 0.001–0.52). In the other 68% ($w_a > 0$) the separation
 1537 condition of Theorem 1(2) is satisfied in *every* round, with margin $\Delta'_{\text{sep}}/2(R'_B + \delta') \geq 2.8$ (mean
 1538 3.2) and δ'/R'_B mean 0.19, max 0.72. The two cases are exhaustive, so the analysis covers all
 1539 measured rounds. Proposition 1(a) verified in the shipped implementation: FoolsGold’s weight
 1540 vector changes by at most 3×10^{-8} (float32 rounding) under NormClip across 1000 synthetic trials
 1541 and live rounds in which 1–2 of 5 clients were genuinely clipped (raw norms up to 24.5 against
 1542 $\tau=5$). Exactly 2 of the 5 clients per round receive zero FoolsGold weight. **Corollary 1 in full:**
 1543 with $D_b := R'_B + \delta'$ bounding every benign residual and $D_a := \Delta'_{\text{sep}} - R'_B - \delta' = (2m - 1)D_b$
 1544 lower-bounding every adversarial one, the Weiszfeld coefficients satisfy $\sum_j 1/d_j \geq n_b/D_b$ and

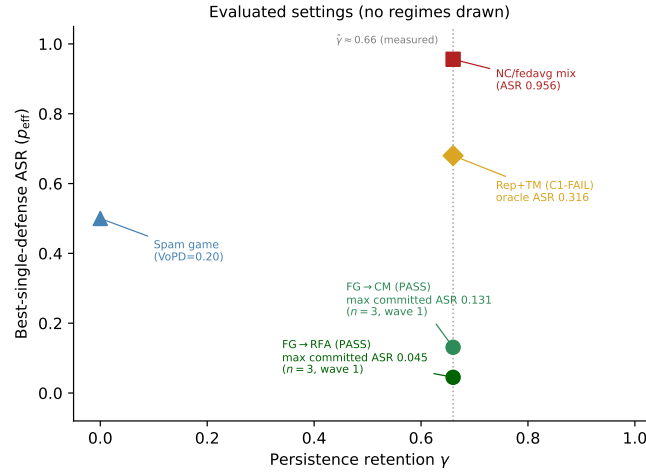


Figure 7: **The evaluated settings, plotted. No regimes, and one line, which is a measurement.** X-axis: persistence γ . Y-axis: best single-defense ASR (p_{eff}). Each marker is a setting we ran and carries its own measured value; the dotted line is the measured FL persistence $\hat{\gamma} \approx 0.66$. **An earlier version shaded four regime bands and drew threshold guides at $\gamma=0.15$, $p_{\text{eff}}=0.3$ and $p_{\text{eff}}=0.6$; none of those numbers was measured, so all six are gone.** Five points cannot locate a boundary, and nothing here is evidence of a phase transition in (γ, p_{eff}) space.

1545 $1/d_a \leq 1/D_a$, so $\Lambda_a \leq (n_a/D_a)/(n_b/D_b + n_a/D_a) = n_a/(n_b(2m-1) + n_a)$. At $m \rightarrow 1$ this
 1546 is $n_a/(n_b + n_a)$, i.e. 40% for $n_a=2$, $n_b=3$; at the measured $m \geq 2.82$ it is $\leq 12.6\%$, and at the
 1547 all-round mean $m = 3.24$, $\leq 10.8\%$.

1548 **Pre-registration provenance.** We distinguish the tiers explicitly. The 18-pair development set is
 1549 where the criterion was *developed*; its predictions were not committed in advance and we report it
 1550 as fit. The 24-pair held-out set was pre-registered in `pre_registration_wave2.md` and committed
 1551 before those experiments ran; it contains no predicted-PASS pair, so it bears only on specificity. The
 1552 out-of-sample PASS predictions (DBA transfer, $N=100$ /ResNet18 scale transfer, composition dom-
 1553 inance) were pre-registered separately in `pre_registration_oos_pass.md`, also committed in ad-
 1554 vance. The 10 three-way predictions were pre-registered in `pre_registration_three_way.md`.
 1555 Two held-out LOW calls, hedged as speculative at pre-registration time, failed; we score
 1556 them as misses rather than dropping them. The prospective suite of §5 was pre-registered
 1557 in `pre_registration_prospective.md` at commit 262cf35, which precedes the first write
 1558 to `results/prospective_suite/`; single-defense baselines for the three new defenses were
 1559 measured *before* the freeze, because C1 and C0 take them as inputs (FLTrust 0.048, Krum
 1560 0.583, Multi-Krum 0.783 max-committed; FedAvg 0.667 scaling / 0.768 pixel, clearing the
 1561 sanity gate). The predicted labels in that file are reproduced verbatim in the PAIRS list of
 1562 `run_prospective_suite.py` and the confusion matrix is recomputed from per-seed ASR in
 1563 `results/prospective_suite/summary.json`, not transcribed.

1564 **Freeze commits and run tallies for the interventional suites.** Every design,
 1565 grid and threshold below was git-committed before the corresponding results direc-
 1566 tory existed, and each hash is verifiable in the repository history. *Dose-response*
 1567 *ladder* (§5): `pre_registration_dose_response.md` at commit e711a95, before
 1568 `results/dose_response/`; 80 runs. *Metric-swap mirrors* (§5): 8 frozen labels at commit
 1569 356897b. *Prospective suite* (§5): commit 262cf35, as above. *Targeted intervention, Modes S*
 1570 *and A* (§4, §5): design and thresholds at commit 5130cec before `results/targeted_dose/`
 1571 existed; 97 runs plus 25 identity runs verified bit-identical per seed (largest deviation 0.00e+0);
 1572 the adversarial coefficient share was verified with no ASR involved, before any run. *Rank-*
 1573 *discordant replication* (§4): `pre_registration_dose_replication.md` at commit 06d9bce
 1574 before `results/dose_replication/` existed; 15 new runs, identity rung imported bit-identical
 1575 from the ladder and verified by `--harness-check`. *Second dataset and architecture* (§4):
 1576 `pre_registration_dose_femnist.md` at commit 478b555 before `results/dose_femnist/`
 1577 existed; 12 new runs, with $\kappa=0$ run in-suite because no EMNIST-byclass ladder exists to
 1578 import from. *Score-only control* (§4, Appendix E): `pre_registration_score_only.md` at
 1579 commit 35788d9 before `results/score_only/` existed; 15 new runs, identity rung imported
 1580 bit-identical and verified by `--harness-check`, and the `cos_krum` corollary checked at one seed
 1581 as pre-registered.